

A DC Composite Optimization via Variable Smoothing for Robust Phase Retrieval with Nonconvex Loss Functions

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Abstract—In this paper, we propose an optimization-based method for robust phase retrieval problem where the goal is to estimate an unknown signal from a quadratic measurement corrupted by outliers. To enhance the robustness of existing optimization models with the ℓ_1 loss function, we propose a generalized model that can handle DC (Difference-of-Convex) loss functions beyond the ℓ_1 loss. We view the cost function of the proposed model as a composition of a DC function with a smooth mapping, and develop a variable smoothing algorithm for minimizing such DC composite functions. At each step of our algorithm, we generate a smooth surrogate function by using the Moreau envelope of each (weakly) convex function in the DC function, and then perform the gradient descent update of the surrogate function. Unlike many existing algorithms for DC problems, the proposed algorithm does not require any inner loop. We also present a convergence analysis in terms of a DC composite critical point for the proposed algorithm. Our numerical experiment demonstrates that the proposed method with DC loss functions is more robust against outliers compared to existing methods with the ℓ_1 loss.

Index Terms—Robust phase retrieval, nonsmooth optimization, DC composite, Moreau envelope, variable smoothing

I. INTRODUCTION

PHASE retrieval is a problem of estimating an original signal $\mathbf{x}^* \in \mathbb{R}^d$ or $-\mathbf{x}^*$ from a quadratic measurement

$$\mathbf{b}^* := [\langle \mathbf{a}_1, \mathbf{x}^* \rangle^2, \langle \mathbf{a}_2, \mathbf{x}^* \rangle^2, \dots, \langle \mathbf{a}_n, \mathbf{x}^* \rangle^2]^T \in \mathbb{R}^n,$$

where $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n \in \mathbb{R}^d$ are known measurement vectors.¹The measurement $\mathbf{b}^*$ can also be expressed as

$$\mathbf{b}^* = (A\mathbf{x}^*) \odot (A\mathbf{x}^*)$$

by using the matrix $A := [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n]^T \in \mathbb{R}^{n \times d}$ and the Hadamard product (i.e., entry-wise product) $\odot$. Phase retrieval

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¹The name *phase retrieval* comes from its applications where the measurement matrix $A := [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n]^T$ and the measurement $\mathbf{b}^* := [|\langle \mathbf{a}_1, \mathbf{x}^* \rangle|^2, |\langle \mathbf{a}_2, \mathbf{x}^* \rangle|^2, \dots, |\langle \mathbf{a}_n, \mathbf{x}^* \rangle|^2]^T \in \mathbb{R}^n$ represent a Fourier-type transform and a phaseless measurement, respectively. (For this reason, A is also often considered as a complex matrix in phase retrieval literature; however, in this paper, we assume A to be real-valued for simplicity as in many previous studies [2]–[6].)

problem arises in various applications including crystallography [7], [8], optics [9], [10], and astronomy [11]. In this paper, we consider a scenario where only the following measurement $\mathbf{b}$, corrupted by outliers, is available:

$$[\mathbf{b}]_i := \begin{cases} \langle \mathbf{a}_i, \mathbf{x}^* \rangle^2 + \varepsilon_i & i \in \mathcal{I}_{\text{in}} \\ \xi_i & i \in \mathcal{I}_{\text{out}}, \end{cases}$$

in which $\mathcal{I}_{\text{out}} \subset \{1, 2, \dots, n\}$ and $\mathcal{I}_{\text{in}} := \{1, 2, \dots, n\} \setminus \mathcal{I}_{\text{out}}$ denote index sets for outliers $\xi_i \in \mathbb{R}$ and inliers respectively, and $\varepsilon_i \in \mathbb{R}$ represents a certain additive noise, e.g., white Gaussian noise. Such a corrupted measurement often appears in many phase retrieval imaging applications [12], for example due to sensor failures and recording errors.

In a case where there is no outlier, i.e., $\mathcal{I}_{\text{out}} = \emptyset$, a variety of phase retrieval methods have been proposed over the decades, including *Gerchberg-Saxton algorithm* [13], *hybrid-input output method* [14], [15], *PhaseLift* [16], and *Wirtinger flow* [17], to name a few. Recently, for the case $\mathcal{I}_{\text{out}} \neq \emptyset$, numerous studies have aimed to develop *robust phase retrieval* methods for achieving high-precision estimation even in the presence of outliers [2]–[6], [12], [18]. Among these, [18] and [5] consider optimization-based methods analogous to the well-known *least absolute deviation method* (see, e.g., [19]) in the field of robust statistics. To be precise, they utilize a solution of the nonconvex optimization model

$$\underset{\mathbf{x} \in \mathbb{R}^d}{\text{minimize}} \Phi_1(\mathbf{x}) := \|(A\mathbf{x}) \odot (A\mathbf{x}) - \mathbf{b}\|_1 = \sum_{i=1}^n |\langle \mathbf{a}_i, \mathbf{x} \rangle^2 - [\mathbf{b}]_i| \quad (1)$$

as an estimated signal of $\mathbf{x}^*$ or $-\mathbf{x}^*$, where the ℓ_1 norm $\|\cdot\|_1 : \mathbb{R}^n \rightarrow \mathbb{R}$, $\mathbf{z} \mapsto \sum_{i=1}^n |z_i|$ serves as a loss function. In addition, [12], [2], and [6] also adopt similar optimization models with the ℓ_1 loss function. More specifically, [12] uses a sparse regularized version of (1) under the sparsity assumption on the target signal $\mathbf{x}^*$; [2] studies a convex relaxation of the model (1); and [6] employs a modified formulation of (1)

$$\underset{\mathbf{x} \in \mathbb{R}^d}{\text{minimize}} \Phi_2(\mathbf{x}) := \sum_{i=1}^n \left| |\langle \mathbf{a}_i, \mathbf{x} \rangle| - \sqrt{[\mathbf{b}]_i} \right|, \quad (2)$$

in which $\langle \mathbf{a}_i, \mathbf{x} \rangle^2$ and $[\mathbf{b}]_i$ are replaced by their square roots $|\langle \mathbf{a}_i, \mathbf{x} \rangle|$ and $\sqrt{[\mathbf{b}]_i}$, respectively. It is reported in [6], [18] that these methods using the ℓ_1 norm achieve superior numerical estimation performance compared to other robust phase retrieval methods [3], [4].

Nevertheless, it is questionable whether the ℓ_1 norm in the model (1) can adequately suppress effects caused by

TABLE I: Examples of DC loss function φ in the model (3)

name	$\varphi(\mathbf{z}) = (f - g)(\mathbf{z})$	$f(\mathbf{z})$	$g(\mathbf{z})$
ℓ_1 norm	$\sum_{i=1}^n \mathbf{z} _i $	$\sum_{i=1}^n \mathbf{z} _i $	0
MCP [23]	$\sum_{i=1}^n r_{\lambda,\beta}(\mathbf{z} _i)^{*a}$ ($\lambda, \beta \in \mathbb{R}_{++}$)	$\lambda \sum_{i=1}^n \mathbf{z} _i $	$\sum_{i=1}^n \hat{r}_{\lambda,\beta}(\mathbf{z} _i)^{*b}$
Capped ℓ_1 [24]	$\sum_{i=1}^n \min\{ \mathbf{z} _i , \beta\}$ ($\beta \in \mathbb{R}_{++}$)	$\sum_{i=1}^n \mathbf{z} _i $	$\sum_{i=1}^n \max\{ \mathbf{z} _i - \beta, 0\}$
Trimmed ℓ_1 [25]	$\sum_{i=K+1}^n \mathbf{z} _{\downarrow i} ^{*c}$ ($0 \leq K < n$)	$\sum_{i=1}^n \mathbf{z} _i $	$\sum_{i=1}^K \mathbf{z} _{\downarrow i} $

$$^{*a} r_{\lambda,\beta}(t) := \begin{cases} \lambda|t| - \frac{t^2}{2\beta} & |t| \leq \beta\lambda, \\ \frac{\beta\lambda^2}{2} & \text{otherwise.} \end{cases}$$

$$^{*b} \hat{r}_{\lambda,\beta}(t) := \begin{cases} \frac{t^2}{2\beta} & |t| \leq \beta\lambda, \\ \lambda|t| - \frac{\beta\lambda^2}{2} & \text{otherwise.} \end{cases}$$

$^{*c} [\mathbf{z}]_{\downarrow i}$ denotes the entry of $\mathbf{z}$ whose absolute value is the i -th largest.

outliers. To explain this, we rewrite the cost function in (1) as $\sum_{i \in \mathcal{I}_{\text{in}}} |\langle \mathbf{a}_i, \mathbf{x} \rangle^2 - \langle \mathbf{a}_i, \mathbf{x}^* \rangle^2 - \varepsilon_i| + \sum_{i \in \mathcal{I}_{\text{out}}} |\langle \mathbf{a}_i, \mathbf{x} \rangle^2 - \xi_i|$. When there are numerous outliers, or when each ξ_i is large, the second summation also becomes large even if $\mathbf{x}$ is close to the target signal $\mathbf{x}^*$ or $-\mathbf{x}^*$. In this situation, solutions of (1) may deviate from the target $\mathbf{x}^*$ or $-\mathbf{x}^*$, from which the performance of the estimation via (1) may deteriorate. Indeed, similar deteriorations caused by the ℓ_1 loss have been reported in robust regression [20], robust matrix factorization [21], and robust tensor recovery [22]. From these observations, the ℓ_1 norm is not necessarily an ideal loss function for achieving a robust estimation against outliers. Hence, it is expected that the estimation performance of (1) can be enhanced by replacing the ℓ_1 norm with a more appropriate robust loss function.

In this paper, in order to adopt more robust loss functions than the ℓ_1 norm, we propose the following generalized model of (1):

$$\text{minimize}_{\mathbf{x} \in \mathbb{R}^d} \Phi_3(\mathbf{x}) := \varphi((A\mathbf{x}) \odot (A\mathbf{x}) - \mathbf{b}), \quad (3)$$

where $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}$ is given as a DC (Difference-of-Convex) function, i.e., φ can be expressed as a difference $f - g$ of two convex functions f and $g : \mathbb{R}^n \rightarrow \mathbb{R}$. The DC loss functions φ include not only the ℓ_1 norm but also nonconvex functions such as the MCP (Minimax-Concave-Penalty) function [20], [23], the capped ℓ_1 norm [24], [26], and the trimmed ℓ_1 norm [25], [27]–[29] to name a few (see Table I for typical DC loss functions and their DC decompositions). Such nonconvex DC functions have been employed as loss functions instead of the ℓ_1 norm in the fields of robust estimation, such as robust regression [20], [30] and robust low rank matrix recovery [26], [31] (see Definition I.1 for intuitive reasons why these nonconvex functions are promising robust loss functions).

Remark I.1 (Robustness of nonconvex DC functions).

- (a) (Upper bounded loss functions) As seen from Table I, the MCP function and the capped ℓ_1 norm are given

respectively by the separable sum of the univariate functions whose outputs are always bounded above by a certain tunable constant. Hence, with a properly tuned constant, these loss functions do not overpenalize large outliers, unlike the ℓ_1 norm.

- (b) (Trimmed ℓ_1 norm) The trimmed ℓ_1 -norm outputs the sum of the smallest $n - K$ absolute values of input vector entries, where K is a tunable parameter. If K is set properly, e.g., as the number of outliers, the trimmed ℓ_1 norm can suppress an influence caused by large outliers. Thus, the trimmed ℓ_1 norm can be an alternative robust loss function.

Although these nonconvex functions are promising loss functions, existing optimization algorithms [2], [5], [6], [12], [18] employed for ℓ_1 loss-based models are not directly applicable to the proposed model (3) with nonconvex φ because these algorithms rely on the convexity of the ℓ_1 norm.

In this paper, we propose an optimization algorithm applicable to the model (3) by exploiting a fact that the cost function in (3) is the composition of a DC function φ with a smooth mapping

$$\mathfrak{S}_{\text{RPR}} : \mathbb{R}^d \rightarrow \mathbb{R}^n, \quad \mathbf{x} \mapsto (A\mathbf{x}) \odot (A\mathbf{x}) - \mathbf{b}. \quad (4)$$

To broaden the applicability of the proposed algorithm (see Definition I.3 (c)), we consider the following optimization problem that includes the model (3) (see Definition I.3 (b)).

Problem I.2 (DC composite-type problem).

$$\text{minimize}_{\mathbf{x} \in \mathbb{R}^d} F(\mathbf{x}) := \underbrace{(f - g)}_{\varphi} \circ \mathfrak{S}(\mathbf{x}), \quad (5)$$

where

- (a) $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $g : \mathbb{R}^n \rightarrow \mathbb{R}$ are
- (i) η_f - and η_g -weakly convex with $\eta_f, \eta_g > 0$, i.e., $f + \frac{\eta_f}{2} \|\cdot\|^2$ and $g + \frac{\eta_g}{2} \|\cdot\|^2$ are convex (we define $\eta := \max\{\eta_f, \eta_g\}$ for convenience),
 - (ii) L_f - and L_g -Lipschitz continuous with $L_f, L_g > 0$, i.e., $|f(\mathbf{x}) - f(\mathbf{y})| \leq L_f \|\mathbf{x} - \mathbf{y}\|$ and $|g(\mathbf{x}) - g(\mathbf{y})| \leq L_g \|\mathbf{x} - \mathbf{y}\|$ hold for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$,
 - (iii) prox-friendly, i.e., their *proximity operators* (see Definition II.5) are available as computable tools (see Definition I.3 (a) for a reason why we assume weak convexity in (i) instead of convexity);
- (b) $\mathfrak{S} : \mathbb{R}^d \rightarrow \mathbb{R}^n$ is differentiable and its Fréchet derivative $D\mathfrak{S} : \mathbb{R}^d \rightarrow \mathbb{R}^{n \times d}$ is $L_{D\mathfrak{S}}$ -Lipschitz continuous (see Notation for the definition of Fréchet derivative);
- (c) F is bounded below, i.e., $\inf_{\mathbf{x} \in \mathbb{R}^d} F(\mathbf{x}) > -\infty$.

Remark I.3 (Applicability of Definition I.2).

- (a) (Functions expressed as $f - g$) All functions φ in Table I admit DC decompositions $\varphi = f - g$ where both f and g are just convex, Lipschitz continuous, and prox-friendly² functions. By virtue of assuming weak convex-

²The proximity operator of every f in Table I is found, e.g., in [32, Exm. 6.8]. On the other hand, the proximity operators of g for the MCP function and the capped ℓ_1 norm can be expressed with those of the univariate prox-friendly functions $\hat{r}_{\lambda,\beta}$ and $\max\{|\cdot| - \beta, 0\}$ [32, Thm. 6.6] (see also [32, Exm. 6.66, Thm. 6.12] and [33] for the detailed expressions of their proximity operators). Lastly, the proximity operator of $g(\mathbf{z}) := \sum_{i=1}^K ||\mathbf{z}|_{\downarrow i}|$ ($\mathbf{z} \in \mathbb{R}^n$) can be computed by a special case of [34, Alg.4].

ity (rather than convexity) for f and g as in (a)(i), we can also employ a wider variety of robust loss functions (or sparsity-promoting functions; see Definition I.3 (c)) as φ beyond those listed in Table I. For example, the *Cauchy loss* (see, e.g., [35]) and the *log-sum penalty* [36] are weakly convex, Lipschitz continuous, and prox-friendly; hence, they can be used as φ in Definition I.2 by setting $f = \varphi$ and $g \equiv 0$.

- (b) (Robust phase retrieval) The model (3) with all φ in Table I is reproduced as a special case of Definition I.2 by setting f and g as in Table I, and $\mathfrak{S} := \mathfrak{S}_{\text{RPR}}$. Indeed, f and g in Table I satisfy the assumptions in Definition I.2 as stated in (a), and $D\mathfrak{S}_{\text{RPR}} : \mathbf{x} \mapsto [2\langle \mathbf{a}_1, \mathbf{x} \rangle \mathbf{a}_1, 2\langle \mathbf{a}_2, \mathbf{x} \rangle \mathbf{a}_2, \dots, 2\langle \mathbf{a}_n, \mathbf{x} \rangle \mathbf{a}_n]^T$ is $(2\sqrt{\sum_{i=1}^n \|\mathbf{a}_i\|^4})$ -Lipschitz continuous (see (30)).
- (c) (Example of applications beyond phase retrieval) DC functions $\varphi = f - g$ in Table I have also been used as regularization functions to promote the sparsity of $\mathfrak{S}(\mathbf{x})$. Hence, Definition I.2 also appears in sparsity-aware applications such as image restoration [37], compressed sensing [38], and cardinality-constrained linear regression [28]. In such applications, optimization models with a sparse regularization term $(f - g) \circ \mathfrak{S}(\mathbf{x})$ are typically formulated as

$$\underset{\mathbf{x} \in \mathbb{R}^d}{\text{minimize}} \quad h(\mathbf{x}) + (f - g) \circ \mathfrak{S}(\mathbf{x}), \quad (6)$$

where $h : \mathbb{R}^d \rightarrow \mathbb{R}$ is differentiable with a Lipschitz continuous gradient and serves as a data fidelity, e.g., least squares. The problem (6) is a special instance of Definition I.2, because the cost function of (6) can be translated into the form $(\hat{f} - \hat{g}) \circ \hat{\mathfrak{S}}(\mathbf{x})$ by introducing $\hat{\mathfrak{S}} : \mathbb{R}^d \rightarrow \mathbb{R}^n \times \mathbb{R} : \mathbf{x} \mapsto [\mathfrak{S}(\mathbf{x})^T, h(\mathbf{x})]^T$, $\hat{f} : \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R} : [\mathbf{z}^T, t]^T \mapsto f(\mathbf{z}) + t$, and $\hat{g} : \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R} : [\mathbf{z}^T, t]^T \mapsto g(\mathbf{z})$. Indeed, if f, g , and $\mathfrak{S}$ satisfy the assumptions in Definition I.2, then $\hat{f}, \hat{g}$, and $\hat{\mathfrak{S}}$ do as well.

The proposed algorithm for DC composite-type problem (Definition I.2) is designed as a gradient descent update of a time-varying smoothed surrogate function of F in (5). With the *Moreau envelopes* (see Definition II.5) ${}^\mu f$ of f and ${}^\mu g$ of g , the proposed surrogate function is given as $({}^\mu f - {}^\mu g) \circ \mathfrak{S}$, where $(\mu_k)_{k=1}^\infty \subset \mathbb{R}$ is a monotonically decreasing sequence of convergence to zero. By utilizing the proximity operators of f and g , the proposed algorithm can be implemented as a single-loop algorithm for Definition I.2 including the model (3). We present an asymptotic convergence analysis (Definition III.7) in the sense of a *DC composite critical point* (see Definition II.3) under Assumption III.2 on the surrogate function. (This assumption is satisfied for the model (3); see Definition III.3 for details.)

Our numerical experiment in scenarios with numerous outliers demonstrates that the proposed method, based on the model (3) with DC loss functions, achieves higher estimation performance than existing state-of-the-art methods [5], [6].

Related works. For Definition I.2, a recently developed *DC composite algorithm* (DCCA) [39] can be used. If an exact solution to a certain subproblem in DCCA is available,

then DCCA has a convergence guarantee in terms of a DC composite critical point. In practice, however, DCCA requires an infinite number of iterations of an inner loop in order to find the exact solution of the subproblem. The convergence analysis of DCCA does not cover realistic cases where only inexact solutions of the subproblem are available. In contrast, the proposed algorithm has a convergence guarantee and does not require infinite iterations of an inner loop.

The proposed algorithm serves as an extension of *variable smoothing-type algorithms* [40], [41], originally developed for a special case $g \equiv 0$ of the problem (6) (more precisely, $\mathfrak{S}$ is assumed to be linear in [40]), where (6) is an instance of Definition I.2 (see Definition I.3 (c)). Note that our extension enables us to cover the model (3) with non-weakly convex DC loss function φ such as the capped ℓ_1 norm and the trimmed ℓ_1 norm.

For a special case $\mathfrak{S} = \text{Id}$ of (6), we also found a similar algorithm [42] to the proposed algorithm in the sense that the Moreau envelopes of f and g are exploited. This algorithm is based on a certain approximate gradient descent method of a smoothed surrogate function $h + {}^\mu f - {}^\mu g$ with fixed $\mu > 0$. Even with such a fixed surrogate function, the algorithm [42] has a convergence guarantee to a DC composite critical point (see [42, Thm. 2]) without requiring any inner loop. However, we have not found yet any extension of the idea in [42] that can handle a nonlinear $\mathfrak{S}$ such as $\mathfrak{S}_{\text{RPR}}$.

Notation. $\mathbb{N}$, $\mathbb{R}$ and $\mathbb{R}_{++}$ denote respectively the sets of all positive integers, all real numbers and all positive real numbers. $\|\cdot\|$ and $\langle \cdot, \cdot \rangle$ are respectively the Euclidean norm and the Euclidean inner product. For subsets S_1, S_2 of Euclidean space, we define $S_1 \pm S_2 := \{\mathbf{v}_1 \pm \mathbf{v}_2 \mid \mathbf{v}_1 \in S_1, \mathbf{v}_2 \in S_2\}$. For $\mathbf{v} \in \mathbb{R}^n$, $[\mathbf{v}]_i \in \mathbb{R}$ stands for the i -th entry. The operator norm for a matrix $X \in \mathbb{R}^{n \times d}$ is defined by $\|X\|_{\text{op}} := \sup_{\|\mathbf{y}\| \leq 1} \|X\mathbf{y}\|$. We use Id to denote the identity mapping. For Euclidean spaces $\mathcal{X}, \mathcal{Y}$ and a continuously differentiable mapping $J : \mathcal{X} \rightarrow \mathcal{Y}$, its Fréchet derivative at $\mathbf{x} \in \mathcal{X}$ is the linear operator $DJ(\mathbf{x}) : \mathcal{X} \rightarrow \mathcal{Y}$ such that $\lim_{\mathbf{h} \rightarrow 0} \frac{J(\mathbf{x} + \mathbf{h}) - J(\mathbf{x}) - DJ(\mathbf{x})[\mathbf{h}]}{\|\mathbf{h}\|} = 0$. (Note that we also regard $DJ(\mathbf{x})$ as a matrix, because every linear operator can be represented by matrix-vector multiplication in finite dimensions.) In particular with $\mathcal{Y} = \mathbb{R}$, $\nabla J : \mathcal{X} \rightarrow \mathcal{X}$ is called the gradient of J if $\nabla J(\mathbf{x}) \in \mathcal{X}$ at $\mathbf{x} \in \mathcal{X}$ satisfies $DJ(\mathbf{x})[\mathbf{v}] = \langle \nabla J(\mathbf{x}), \mathbf{v} \rangle$ ($\mathbf{v} \in \mathcal{X}$). For a point sequence $(\mathbf{p}_k)_{k=1}^\infty \subset \mathcal{X}$, we define its *outer limit* as

$$\text{Limsup}_{k \rightarrow \infty} \mathbf{p}_k := \{\mathbf{p} \in \mathcal{X} \mid \mathbf{p} \text{ is a cluster point of } (\mathbf{p}_k)_{k=1}^\infty\},$$

where the outer limit is originally defined for set sequences (see, e.g., [43, Def. 4.1]), but we only use the outer limit for point sequences (i.e., the outer limit for sequences of singletons).

II. PRELIMINARY

As an extension of the subdifferential of convex functions, we use the following subdifferential of nonconvex functions (see, e.g., a recent survey [44] for readers who are unfamiliar with nonsmooth analysis).

Definition II.1 (Subdifferential [43, Def. 8.3]). For a function $\psi : \mathbb{R}^N \rightarrow \mathbb{R}$, the *limiting (or general) subdifferential* of ψ at $\bar{x} \in \mathbb{R}^N$ is defined by

$$\partial_L \psi(\bar{x}) := \left\{ v \in \mathbb{R}^N \left| \begin{array}{l} \exists (x_k)_{k=1}^\infty \rightarrow \bar{x}, \exists v_k \in \partial_F \psi(x_k) \ (k \in \mathbb{N}) \\ \text{s.t. } (v_k)_{k=1}^\infty \rightarrow v \text{ and } \psi(x_k) \rightarrow \psi(\bar{x}) \end{array} \right. \right\}.$$

Here, $\partial_F \psi(\hat{x}) \subset \mathbb{R}^N$ denotes the *Fréchet (or regular) subdifferential* at $\hat{x} \in \mathbb{R}^N$, and it is the set of all vectors $w \in \mathbb{R}^N$ such that

$$\lim_{\delta \searrow 0} \inf_{0 < \|x - \hat{x}\| < \delta} \frac{\psi(x) - \psi(\hat{x}) - \langle w, x - \hat{x} \rangle}{\|x - \hat{x}\|} \geq 0.$$

From the definition, $\partial_L \psi(\bar{x}) \supset \partial_F \psi(\bar{x})$ obviously holds. If ψ is convex, the limiting subdifferential is equivalent to the convex subdifferential [43, Prop. 8.12]. Furthermore, if ψ is continuously differentiable on a neighborhood of $\bar{x}$, $\partial_L \psi(\bar{x}) = \{\nabla \psi(\bar{x})\}$ holds [43, Exe. 8.8 (b)].

Fact II.2 ([45, (Step1) of Proof of Lem. 2.4], [43, Cor. 8.11]). Consider Definition I.2. Then, for any $x \in \mathbb{R}^d$, we have³

$$\partial_L(f \circ \mathfrak{S})(x) = \partial_F(f \circ \mathfrak{S})(x), \quad \partial_L(g \circ \mathfrak{S})(x) = \partial_F(g \circ \mathfrak{S})(x). \quad (7)$$

By Definition II.2, useful facts for limiting and Fréchet subdifferentials (see, e.g., [43, Ch. 8]) are applicable to $f \circ \mathfrak{S}$ and $g \circ \mathfrak{S}$.

Unfortunately, finding a global minimizer of Definition I.2 is not realistic due to the severe nonconvexity of F . Instead, in this paper, we focus on finding a DC composite critical point defined, with the limiting subdifferentials, as follows.

Definition II.3 (DC composite critical point for Definition I.2 [39, Def. 1.1]). We call $x^* \in \mathbb{R}^d$ a *DC composite critical point* for Definition I.2 if

$$\partial_L(f \circ \mathfrak{S})(x^*) - \partial_L(g \circ \mathfrak{S})(x^*) \ni 0, \quad (8)$$

or equivalently,

$$\partial_L(f \circ \mathfrak{S})(x^*) \cap \partial_L(g \circ \mathfrak{S})(x^*) \neq \emptyset. \quad (9)$$

(More precisely, [39, Def. 1.1] employs the condition obtained by replacing ∂_L in (9) with ∂_F ; however, Definition II.2 shows that this condition coincides with (9) in our setting.)

Lemma II.4 (Local optimality implies DC composite criticality). Let $x^* \in \mathbb{R}^d$ be a local minimizer of F in Definition I.2. Then, x^* is a DC composite critical point for Definition I.2.

Proof. See Appendix A. $\square$

From Definition II.4, being a DC composite critical point is a necessary condition for being a local minimizer. In a case where $f \circ \mathfrak{S}$ and $g \circ \mathfrak{S}$ are convex, finding such a DC composite critical point has been used as an acceptable goal in many DC optimization literature [28], [42], [46], [47]. Even when $f \circ \mathfrak{S}$

³Equation (7) is obtained by combining [45, (Step1) of Proof of Lem.2.4] and [43, Cor.8.11] as below. Since $f \circ \mathfrak{S}$ and $g \circ \mathfrak{S}$ have a property called *subdifferential regularity* (see [43, Def. 7.25]) by [45, (Step1) of Proof of Lem. 2.4], we see from [43, Cor. 8.11] that (7) holds if $\partial_L(f \circ \mathfrak{S})(x) \neq \emptyset$ and $\partial_L(g \circ \mathfrak{S})(x) \neq \emptyset$. In a case where $\partial_L(f \circ \mathfrak{S})(x) = \emptyset$ or $\partial_L(g \circ \mathfrak{S})(x) = \emptyset$, (7) trivially holds from $\partial_L(f \circ \mathfrak{S})(x) \supset \partial_F(f \circ \mathfrak{S})(x)$ and $\partial_L(g \circ \mathfrak{S})(x) \supset \partial_F(g \circ \mathfrak{S})(x)$.

and $g \circ \mathfrak{S}$ are not convex, DC composite critical points are adopted as the target of DC composite algorithm in [39].

The Moreau envelope plays an important role in this paper for designing the proposed algorithm.

Definition II.5 (Moreau envelope, proximity operator [40]). Let $\psi : \mathbb{R}^n \rightarrow \mathbb{R}$ be an η_ψ -weakly convex function with $\eta_\psi > 0$. Its Moreau envelope and proximity operator at $\bar{z} \in \mathbb{R}^n$ with $\mu \in (0, \eta_\psi^{-1})$ are respectively defined as

$$\begin{aligned} {}^\mu \psi(\bar{z}) &:= \min_{z \in \mathbb{R}^n} \left\{ \psi(z) + \frac{1}{2\mu} \|z - \bar{z}\|^2 \right\}, \\ \text{Prox}_{\mu\psi}(\bar{z}) &:= \operatorname{argmin}_{z \in \mathbb{R}^n} \left\{ \psi(z) + \frac{1}{2\mu} \|z - \bar{z}\|^2 \right\}, \end{aligned}$$

where $\text{Prox}_{\mu\psi}$ is single-valued due to the strong convexity of $\psi + (2\mu)^{-1} \|\cdot - \bar{z}\|^2$.

The Moreau envelope ${}^\mu \psi$ serves as a smoothed surrogate function of ψ because of the next properties.

Fact II.6 (Properties of Moreau envelope). Let $\psi : \mathbb{R}^n \rightarrow \mathbb{R}$ be an η_ψ -weakly convex function with $\eta_\psi > 0$. For $\mu \in (0, \eta_\psi^{-1})$, the following hold.

- (a) [43, Thm. 1.25] ($z \in \mathbb{R}^n$) $\lim_{\mu \searrow 0} {}^\mu \psi(z) = \psi(z)$.
- (b) [48, Cor. 3.4] ${}^\mu \psi$ is continuously differentiable with ($z \in \mathbb{R}^n$) $\nabla {}^\mu \psi(z) = \mu^{-1}(z - \text{Prox}_{\mu\psi}(z))$.
- (c) [48, Cor. 3.4] $\nabla {}^\mu \psi$ is $L_{\nabla {}^\mu \psi}$ -Lipschitz continuous with $L_{\nabla {}^\mu \psi} := \max\{\mu^{-1}, \frac{\eta_\psi}{1 - \eta_\psi \mu}\}$.

Note that for f and g in Definition I.2, we can compute $\nabla {}^\mu f$ and $\nabla {}^\mu g$ in closed-forms because these functions are assumed to be prox-friendly (see Definition I.2 (a)(iii)).

III. VARIABLE SMOOTHING ALGORITHM FOR DC COMPOSITE PROBLEM

A. Design of Smooth Surrogate Function

In our algorithm, we use the smooth surrogate function

$$F^{(\mu)} := ({}^\mu f - {}^\mu g) \circ \mathfrak{S} \quad (\mu \in (0, \eta^{-1})) \quad (10)$$

of F in place of the direct utilization of the nonsmooth function F . The next theorem suggests how to find a DC composite critical point in (8) using the surrogate function $F^{(\mu)}$.

Theorem III.1 (DC gradient sub-consistency). Suppose that a positive sequence $(\mu_k)_{k=1}^\infty \subset (0, \eta^{-1})$ converges to 0. For $F_k := F^{(\mu_k)}$ ($k \in \mathbb{N}$) with (10) and any convergent sequence $(x_k)_{k=1}^\infty \subset \mathbb{R}^d \rightarrow \exists \bar{x} \in \mathbb{R}^d$, the following hold:

(a)

$$\limsup_{k \rightarrow \infty} \nabla F_k(x_k) \subset \partial_L(f \circ \mathfrak{S})(\bar{x}) - \partial_L(g \circ \mathfrak{S})(\bar{x}).$$

(b)

$$\begin{aligned} \text{dist}(\mathbf{0}, \partial_L(f \circ \mathfrak{S})(\bar{x}) - \partial_L(g \circ \mathfrak{S})(\bar{x})) \\ \leq \liminf_{k \rightarrow \infty} \|\nabla F_k(x_k)\|, \end{aligned} \quad (11)$$

where $\text{dist}(v, S) := \inf_{w \in S} \|v - w\|$ stands for the distance between a point $v \in \mathbb{R}^d$ and a set $S \subset \mathbb{R}^d$.

Proof. See Appendix B. $\square$

Definition III.1 (b) implies that $\bar{x}$ is a DC composite critical point in the sense of (8) if the right hand side of (11) is zero. Hence, our goal of finding a DC composite critical point of Definition I.2 is reduced to designing an algorithm to generate a point sequence $(\mathbf{x}_k)_{k=1}^\infty$ such that $\liminf_{k \rightarrow \infty} \|\nabla F_k(\mathbf{x}_k)\| = 0$. To design such an algorithm, we introduce the following assumption. Note that this assumption holds for the model (3) as will be stated in Definition III.3.

Assumption III.2 (Descent assumption). For F in Definition I.2, consider $F^{(\cdot)}$ in (10). There exist $\varpi_1, \varpi_2 \in \mathbb{R}_{++}$ such that the following inequality holds for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$ and $\mu \in (0, (2\eta)^{-1}]$:

$$F^{(\mu)}(\mathbf{y}) \leq F^{(\mu)}(\mathbf{x}) + \langle \nabla F^{(\mu)}(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \frac{\kappa_\mu}{2} \|\mathbf{y} - \mathbf{x}\|^2, \quad (12)$$

where $\kappa_\mu := \varpi_1 + \varpi_2 \mu^{-1}$. (Note: we can implement the proposed algorithm, illustrated in Section III-B, without any knowledge on the value of ϖ_1 and ϖ_2 .)

From the descent lemma (see, e.g., [32, Lem. 5.7]), Assumption III.2 is satisfied if $\nabla F^{(\mu)}$ is Lipschitz continuous with a Lipschitz constant $\varpi_1 + \varpi_2 \mu^{-1}$. By exploiting this standard result, Definition III.3 (b) below presents a sufficient condition for Assumption III.2. However, this sufficient condition can not be applied to the proposed model (3) because $\mathfrak{S}_{\text{RPR}}$ in (4) is not Lipschitz continuous. Instead, we show directly that the model (3) with φ in Table I satisfies Assumption III.2 in Definition III.3 (a).

Proposition III.3 (Sufficient conditions for Assumption III.2).

- (a) For the model (3) with $\varphi = f - g$ chosen from Table I, $F := \Phi_3 = (f - g) \circ \mathfrak{S}_{\text{RPR}}$ with $\mathfrak{S}_{\text{RPR}}$ in (4) satisfies Assumption III.2 with

$$\begin{aligned} \kappa_\mu := 2L_g \sqrt{\sum_{i=1}^n \|\mathbf{a}_i\|^4} + 6\lambda \max_{1 \leq i \leq n} \|\mathbf{a}_i\|^2 \\ + 4 \left(\max_{1 \leq i \leq n} \|\mathbf{a}_i\|^2 \|\mathbf{b}_i\| \right) \mu^{-1}, \end{aligned}$$

where $\lambda > 0$ is a parameter of the MCP function, and λ is understood as 1 for the other φ in Table I.

- (b) If $\mathfrak{S}$ is $L_{\mathfrak{S}}$ -Lipschitz continuous, then Assumption III.2 holds with $\kappa_\mu := L_{\mathfrak{D}}(L_f + L_g) + 2L_{\mathfrak{S}}^2 \mu^{-1}$. (This is a variant of [45, Prop. 4.2 (a)].)
- (c) Consider the cost function $h + F := h + (f - g) \circ \mathfrak{S}$ of the problem (6) and its reformulation $\hat{F} := (\hat{f} - \hat{g}) \circ \hat{\mathfrak{S}}$ in Definition I.3 (c). If $F^{(\cdot)}$ satisfies Assumption III.2 with κ_μ , then $\hat{F}^{(\cdot)}$ also satisfies Assumption III.2 with $\hat{\kappa}_\mu := L_{\nabla h} + \kappa_\mu$ with a Lipschitz constant $L_{\nabla h} > 0$ of ∇h .

Proof. See Appendix C. $\square$

B. Proposed Algorithm and Its Convergence Analysis

We propose Algorithm 1 based on the gradient descent method of the smoothed surrogate function $F_k := F^{(\mu_k)}$.

Algorithm 1 Variable smoothing algorithm for DC composite type problem (Definition I.2)

Input: $\mathbf{x}_1 \in \mathbb{R}^d, (\mu_k)_{k=1}^\infty \subset (0, (2\eta)^{-1}]$ enjoying (13).

- 1: **for** $k = 1, 2, 3, \dots$ **do**
 - 2: Set $F_k := F^{(\mu_k)} = (\mu_k f - \mu_k g) \circ \mathfrak{S}$
 - 3: Obtain γ_k by Algorithm 2
 - 4: $\mathbf{x}_{k+1} \leftarrow \mathbf{x}_k - \gamma_k \nabla F_k(\mathbf{x}_k)$
 - 5: **end for**
-

Algorithm 2 Backtracking algorithm to find γ_k

Input: $\gamma_{\text{init},k} \in \mathbb{R}_{++}$ (see Definition III.6 for its choices), $\rho \in (0, 1), c \in (0, 1)$

- 1: $\tilde{\gamma} \leftarrow \gamma_{\text{init},k}$
- 2: **while** the condition (14) with $\gamma := \tilde{\gamma}$ is false **do**
- 3: $\tilde{\gamma} \leftarrow \rho \tilde{\gamma}$
- 4: **end while**

Output: $\gamma_k := \tilde{\gamma}$

We design $(\mu_k)_{k=1}^\infty \subset (0, (2\eta)^{-1}]$ to satisfy the following condition (introduced in [45]) so as to establish a convergence analysis of Algorithm 1:

$$\begin{cases} \text{(i) } \lim_{k \rightarrow \infty} \mu_k = 0, & \text{(ii) } \sum_{k=1}^\infty \mu_k = \infty, \\ \text{(iii) } (\forall k \in \mathbb{N}) \mu_k \geq \mu_{k+1}. \end{cases} \quad (13)$$

For example, $\mu_k := (2\eta)^{-1} k^{-\frac{1}{\alpha}}$ with $\alpha \geq 1$ enjoys the condition (13) ($\alpha = 3$ is reported to be an appropriate value for a reasonable convergence rate of a special case of Algorithm 1 with $g \equiv 0$ [40], [45]).

We employ Algorithm 2 in order to obtain a stepsize γ_k enjoying the following *Armijo condition* with $\gamma := \gamma_k$:

$$F_k(\mathbf{x}_k - \gamma \nabla F_k(\mathbf{x}_k)) \leq F_k(\mathbf{x}_k) - c\gamma \|\nabla F_k(\mathbf{x}_k)\|^2. \quad (14)$$

Algorithm 2 is called the *backtracking algorithm*, and it has been utilized as a standard stepsize selection for smooth optimization (see, e.g., [49]). The while-loop in Algorithm 2 terminates after a finite number of iterations as follows.

Lemma III.4 (Properties of Algorithm 2). Consider Definition I.2 under Assumption III.2, and Algorithm 1 with arbitrary inputs $\mathbf{x}_1 \in \mathbb{R}^d$ and $(\mu_k)_{k=1}^\infty \subset (0, (2\eta)^{-1}]$. With any inputs $(\gamma_{\text{init},k}, \rho, c) \in \mathbb{R}_{++} \times (0, 1) \times (0, 1)$, Algorithm 2 for estimating γ_k satisfies the following properties:

- (a) Algorithm 2 outputs a stepsize γ_k enjoying the Armijo condition (14) with $\gamma := \gamma_k$ and

$$\gamma_k \geq \min\{\gamma_{\text{init},k}, 2(1-c)\kappa_{\mu_k}^{-1}\rho\}$$

(see Assumption III.2 for the definition of κ_{μ_k}).

- (b) The while-loop in Algorithm 2 is guaranteed to terminate after at most $\max\{0, \lceil \log_\rho(2(1-c)\kappa_{\mu_k}^{-1}\gamma_{\text{init},k}^{-1}) \rceil\}$ iterations, where $\lceil \cdot \rceil$ denotes the ceiling function.

Proof. For any $\gamma \in (0, 2(1-c)\kappa_{\mu_k}^{-1})$, it follows from (12) with $(\mathbf{x}, \mathbf{y}, \mu) := (\mathbf{x}_k, \mathbf{x}_k - \gamma \nabla F_k(\mathbf{x}_k), \mu_k)$ that $F_k(\mathbf{x}_k - \gamma \nabla F_k(\mathbf{x}_k)) \leq F_k(\mathbf{x}_k) + \gamma(2^{-1}\gamma\kappa_{\mu_k} - 1)\|\nabla F_k(\mathbf{x}_k)\|^2 \leq F_k(\mathbf{x}_k) - c\gamma\|\nabla F_k(\mathbf{x}_k)\|^2$. Hence, Algorithm 2 terminates when γ_k becomes less than $2(1-c)\kappa_{\mu_k}^{-1}$ at the latest, which leads to both (a) and (b). $\square$

For our convergence analysis, we choose initial guesses $(\gamma_{\text{init},k})_{k=1}^\infty$ of Algorithm 2 enjoying the next assumption.

Assumption III.5 (Condition for initial guesses $(\gamma_{\text{init},k})_{k=1}^\infty$). Consider Definition I.2 under Assumption III.2. For an input $(\mu_k)_{k=1}^\infty \subset (0, (2\eta)^{-1}]$ of Algorithm 1 and initial guesses $(\gamma_{\text{init},k})_{k=1}^\infty \subset \mathbb{R}_{++}$ in Algorithm 2, the following holds:

$$(\exists \delta > 0, \forall k \in \mathbb{N}) \quad \gamma_{\text{init},k} \geq \delta \kappa_{\mu_k}^{-1}, \quad (15)$$

where κ_{μ_k} is given in Assumption III.2. (Note: any knowledge on the value of δ is not required in Algorithms 1 and 2.)

Example III.6 ($(\gamma_{\text{init},k})_{k=1}^\infty$ achieving Definition III.5). The following choices of initial guesses $(\gamma_{\text{init},k})_{k=1}^\infty$ achieve Definition III.5 (see Appendix D for the proof).

- (a) (Constant multiple of $\kappa_{\mu_k}^{-1}$) We can choose $\gamma_{\text{init},k} := 2(1-c)\kappa_{\mu_k}^{-1}$ ($k \in \mathbb{N}$) in a case where the value κ_{μ_k} is known (see Definition III.3). Algorithm 2 with this $\gamma_{\text{init},k}$ does not execute the while-loop and outputs $\gamma_k := \gamma_{\text{init},k}$ because $\gamma_{\text{init},k}$ is already small enough to satisfy the Armijo condition (14) (see Definition III.4 (b)).
- (b) (Constant value) We can also choose a constant value $\gamma_{\text{init}} \in \mathbb{R}_{++}$ for $\gamma_{\text{init},k}$. With this choice, the stepsize γ_k is selected adaptively through the while-loop in Algorithm 2. In practice, the resulting stepsize tends to be larger than $2(1-c)\kappa_{\mu_k}^{-1}$. Consequently, compared with the choice $\gamma_{\text{init},k} := 2(1-c)\kappa_{\mu_k}^{-1}$ described in (a), this strategy may lead to faster convergence of Algorithm 1.
- (c) (Stepsize used in previous iteration) We can also use $\gamma_{\text{init},k} := \gamma_{k-1}$, that is, the stepsize used in the $(k-1)$ -th iteration of Algorithm 1, where $\gamma_0 \in \mathbb{R}_{++}$ is a given constant. With this choice, the while-loop in Algorithm 2 may terminate in fewer iterations than in the case $\gamma_{\text{init},k} := \gamma_{\text{init}}$ in (b). In our experiment in Section IV, we adopted this initial guess because it empirically yields shorter convergence time of Algorithm 1 than other choices of $\gamma_{\text{init},k}$ in (a) and (b).

Under Assumption III.2 and Definition III.5, we present below a convergence theorem for Algorithm 1.

Theorem III.7 (Convergence theorem). Consider Definition I.2 under Assumption III.2. Let $(\mu_k)_{k=1}^\infty \subset (0, (2\eta)^{-1}]$ and $(\gamma_{\text{init},k})_{k=1}^\infty \subset \mathbb{R}_{++}$ satisfy (13) and Definition III.5, while the remaining inputs $(\mathbf{x}_1, \rho, c) \in \mathbb{R}^d \times (0, 1) \times (0, 1)$ of Algorithms 1 and 2 are arbitrarily chosen. For the function sequence $(F_k)_{k=1}^\infty$ and the point sequence $(\mathbf{x}_k)_{k=1}^\infty$ produced by Algorithm 1, the following hold:

- (a) For any $\underline{k}, \bar{k} \in \mathbb{N}$ such that $\underline{k} \leq \bar{k}$, we have

$$\min_{\underline{k} \leq k \leq \bar{k}} \|\nabla F_k(\mathbf{x}_k)\| \leq \sqrt{\frac{C}{\sum_{k=\underline{k}}^{\bar{k}} \mu_k}}, \quad (16)$$

where $C \in \mathbb{R}_{++}$ is a constant.

- (b)

$$\liminf_{k \rightarrow \infty} \|\nabla F_k(\mathbf{x}_k)\| = 0. \quad (17)$$

- (c) We can choose a subsequence $(\mathbf{x}_{m(l)})_{l=1}^\infty$ such that $\lim_{l \rightarrow \infty} \|\nabla F_{m(l)}(\mathbf{x}_{m(l)})\| = 0$, where $m : \mathbb{N} \rightarrow \mathbb{N}$ is monotonically increasing. Moreover, every cluster

point of $(\mathbf{x}_{m(l)})_{l=1}^\infty$ is a DC composite critical point of Definition I.2.

Proof. See Appendix E. $\square$

IV. NUMERICAL EXPERIMENT

We conducted numerical experiments in order to evaluate estimation performance of our robust phase retrieval method based on the proposed model (3) and Algorithm 1. In our experiments, we adopted the capped ℓ_1 norm and the trimmed ℓ_1 norm as the DC loss function φ in the model (3) (see Table I for the expressions of φ), where several choices of parameters β and K were tested. In Algorithm 1, we used $\mu_k := k^{-1/3}$ ($k \in \mathbb{N}$) for the parameters of the Moreau envelope. To compute the stepsize γ_k via Algorithm 2, we used $(\rho, c) := (0.8, 0.0001)$, which is a typical choice in smooth optimization [49]. As stated in Definition III.6 (c), we employed $\gamma_{\text{init},k} := \gamma_{k-1}$ ($k \in \mathbb{N}$), where γ_0 was set to $\max\{1, \|\nabla F_1(\mathbf{x}_1)\|^{-1}\}$.

For comparison, we also employed state-of-the-art existing methods [5], [6] based on the ℓ_1 loss function. The method in [5] applies the *inexact proximal linear (IPL) algorithm* to the ℓ_1 loss-based model (1). In IPL, the $(k+1)$ -th estimate $\mathbf{x}_{k+1} \in \mathbb{R}^d$ is obtained as an inexact solution of a subproblem

$$\underset{\mathbf{x} \in \mathbb{R}^d}{\text{minimize}} \quad \|\mathfrak{S}_{\text{RPR}}(\mathbf{x}_k) + \text{D}\mathfrak{S}_{\text{RPR}}(\mathbf{x}_k)[\mathbf{x} - \mathbf{x}_k]\|_1 + \frac{1}{2t} \|\mathbf{x} - \mathbf{x}_k\|^2, \quad (18)$$

which is derived by linearizing $\mathfrak{S}_{\text{RPR}}$ in (1) at $\mathbf{x}_k$ and adding a quadratic term with $t \in \mathbb{R}_{++}$. This subproblem is solved by *fast iterative shrinkage-thresholding algorithm (FISTA)* with two possible stopping criteria named (LACC) and (HACC) (see [5, Alg. 2]). In our experiments, we adopted (HACC) because [5, Fig. 2] demonstrates that IPL with (HACC) empirically yields a slightly higher *success rate* than IPL with (LACC) (for the definition of success rate, see the sentence just after (19)). To implement IPL, we used the code released by the author.⁴ The other competing method [6] applies *robust alternating minimization (Robust-AM)* to the modified ℓ_1 loss-based model (2). Robust-AM is derived as a Gauss-Newton method for (2), where its estimate sequence is iteratively updated by

$$\mathbf{x}_{k+1} \in \underset{\mathbf{x} \in \mathbb{R}^d}{\text{argmin}} \quad \sum_{i=1}^n \left| \langle \mathbf{a}_i, \mathbf{x} \rangle - \text{sign}(\langle \mathbf{a}_i, \mathbf{x}_k \rangle) \sqrt{|\langle \mathbf{b} \rangle_i|} \right|.$$

For solving this subproblem, two methods, *ADMM-LAD* and *ADMM-LP*, are introduced in [6]. Robust-AM with ADMM-LAD is reported to achieve almost the same success rate as one with ADMM-LP while exhibiting significantly faster convergence (see [6, Fig. 2]). Hence, we employed ADMM-LAD in our experiments, as in the main experiments in [6]. We implemented Robust-AM by using the code provided by the author.⁵

For the initial point of Algorithm 1, IPL, and Robust-AM, we generated common $\mathbf{x}_1 \in \mathbb{R}^d$ by an existing initialization

⁴The code of IPL can be found in “<https://github.com/zhengzhongpku/IPL-code-share>”.

⁵We received the code of Robust-AM directly from Mr. Kim (The Ohio State University) who is the first author of [6].

method [18, Alg. 3], which was also used in the experiment in [5]. (This initialization is also mentioned in [6] as being consistent with its convergence analysis [6, Thm. 4.1].) We terminated each algorithm when one of the following conditions was met: (i) the relative change in the the cost function value satisfied $\frac{|\Phi_j(\mathbf{x}_k) - \Phi_j(\mathbf{x}_{k-1})|}{|\Phi_j(\mathbf{x}_{k-1})|} < 10^{-7}$ ($j = 1, 2, 3$) where $\mathbf{x}_k$ is the k -th estimate generated by each algorithm, and Φ_1 , Φ_2 and Φ_3 is the cost functions in (1), (2), and (3), respectively, used in IPL, Robust-AM, and Algorithm 1; (ii) the number of iterations reached to 10000; (iii) the running CPU time exceeded 30 seconds. All experiments were performed by MATLAB on MacBook Pro (Apple M3, 16GB).

The problem settings, partially inspired by [5], are as follows. We drew each entry of $A \in \mathbb{R}^{n \times d}$ from the normal distribution $\mathcal{N}(0, 1)$. Each entry of the target signal $\mathbf{x}^* \in \mathbb{R}^d$ was chosen from 1 or -1 with a probability of 0.5 respectively. The additive noise $\varepsilon_i \in \mathbb{R}$ was generated by $\mathcal{N}(0, 10^{-6})$. The index set $\mathcal{I}_{\text{out}}$ was selected uniformly at random with fixed cardinality $\#\mathcal{I}_{\text{out}}$. Here, let $p_{\text{fail}} := \#\mathcal{I}_{\text{out}}/n$ denote the proportion of outliers. Each outlier value $\xi_i \in \mathbb{R}$ ($i \in \mathcal{I}_{\text{out}}$) was generated from (i) the (absolute) Cauchy distribution or (ii) the uniform distribution. More specifically, each ξ_i was given by (i) $\xi_i := s\mathcal{M}\tan(0.5\pi u_i)$ or (ii) $\xi_i := s\mathcal{M}u_i$, where u_i was drawn from the uniform distribution of $[0, 1]$, $\mathcal{M} := \max_{1 \leq i \leq n} \langle \mathbf{a}_i, \mathbf{x}^* \rangle^2$ was a constant, and a parameter $s \in \mathbb{R}_{++}$ was used to control the scale of outliers. (Since the results were similar for both types of outliers, we present only the results for Cauchy outliers in this section, while those for uniformly distributed outliers are provided in the supplementary material.) For every fixed $d, n, \#\mathcal{I}_{\text{out}}$, and s , we performed estimation on 50 random problem instances obtained by varying $\mathbf{x}^*$, A , ε_i , u_i , and elements of $\mathcal{I}_{\text{out}}$. We judged that an estimation succeeded if the relative error at the final estimate $\mathbf{x}^\diamond \in \mathbb{R}^d$ achieved

$$\min\{\|\mathbf{x}^* - \mathbf{x}^\diamond\|, \|\mathbf{x}^* + \mathbf{x}^\diamond\|\} / \|\mathbf{x}^*\| < 10^{-3}. \quad (19)$$

As in [5], we used an estimation performance criterion called “success rate” that is the percentage of the successful estimation out of 50 estimations.

In the first experiment, we evaluated the success rate of each method for $n/d \in \{5, 10, 15, 20\}$ and $p_{\text{fail}} \in \{0.1 + 0.05j \mid j \in \{0, 1, \dots, 10\}\}$. According to the similar experiments in [5], [6], we employed $d \in \{100, 500\}$. We fixed the parameter s at 1.

Fig. 1 and Fig. 2 show the results for the case $d = 100$ and for the case $d = 500$, respectively. In these figures, pixels with 100% success rate are marked by red pentagrams, and are hereafter referred to as *successful pixels*. Fig. 1 and Fig. 2 imply that the proposed method with the capped ℓ_1 norm and the trimmed ℓ_1 norm tends to achieve relatively high success rate especially in upper region of the figures where p_{fail} is large. More precisely, except for the capped ℓ_1 with $(d, \beta) = (100, 10000)$ and $(500, 100)$, all the proposed methods have more successful pixels than the existing methods in the region $p_{\text{fail}} \geq 0.35$.

In particular, we observe that the capped ℓ_1 with properly chosen β allows for more successful estimations in a wider region than existing methods. Specifically, in the case $(d, \beta) =$

$(100, 100)$ (resp. $(500, 1000)$), the set of successful pixels for the capped ℓ_1 contains those for the existing methods and 3 (resp. 8) additional pixels. On the other hand, we found that the trimmed ℓ_1 yields a larger number of successful pixels than existing methods for all tested values of K/n .⁶ These results above are consistent with the intuition in Definition I.1, where DC loss functions are expected to be robust against outliers.

To examine how appropriate choices of β and K change with different scales s of outliers, we conducted the second experiment, where s varied in $\{1, 10, 100, 1000\}$. We fixed (d, p_{fail}) at $(500, 0.4)$ in this experiment.

Fig. 3 shows the success rate of each method for the cases $n/d = 10$ and $n/d = 20$. When $n/d = 10$, Fig. 3 demonstrates that the best performance of the capped ℓ_1 is obtained with $\beta = 1000$ for $s \in \{1, 10\}$, $\beta = 10000$ for $s = 100$, and $\beta = 100000$ for $s = 1000$, which indicates that larger values of β are appropriate for large outliers. In contrast, for the trimmed ℓ_1 , $K/n = 0.4$ consistently yields high success rate across all values of s , which means that K such that $K/n = p_{\text{fail}}$ is appropriate. For the case $n/d = 20$ where a sufficiently large number of measurements is available, the proposed method performs well across a relatively wide range of β and K . (The design of practical parameter selection strategies is beyond the scope of this paper and will be investigated in future work.)

Lastly, we present, in Table II, the running CPU time and the number of iterations of each methods for the representative case $(d, p_{\text{fail}}, s) = (500, 0.35, 1)$. Table II demonstrates that the proposed method consistently requires less running time than IPL. A possible reason for this is that IPL requires solving subproblems (18) in each iteration, resulting in a higher computational cost per iteration. Indeed, Table II also shows that the proposed method has a substantially lower per-iteration time than IPL.

V. CONCLUSION

We proposed the optimization model (3) with DC loss functions for robust phase retrieval. For DC composite-type problem (Definition I.2) including the proposed model, we designed a variable smoothing algorithm (Algorithm 1) with a convergence guarantee in terms of a DC composite critical point. The proposed algorithm was designed to find a DC composite critical point by generating the sequence of points at which the gradient of the smooth surrogate function approaches zero. Through numerical experiments, we demonstrated the robustness of the proposed model, investigated the relationship between the scale or number of outliers and the appropriate values of β and K , and showed the computational efficiency of the proposed algorithm.

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We thank Mr. Kim (The Ohio State University), the first author of [6], for kindly sharing the code of Robust-AM.

⁶However, we also see that the large K results in low success rate when the number n of measurements is small (see the lower-left region of Fig. 1 (h) and Fig. 2 (g)). This is probably because the trimmed ℓ_1 with a large K ignores not only outliers, but also many inliers, thereby excessively reducing the number of effective inlier measurements available for estimation, especially when the number n of measurements is small.

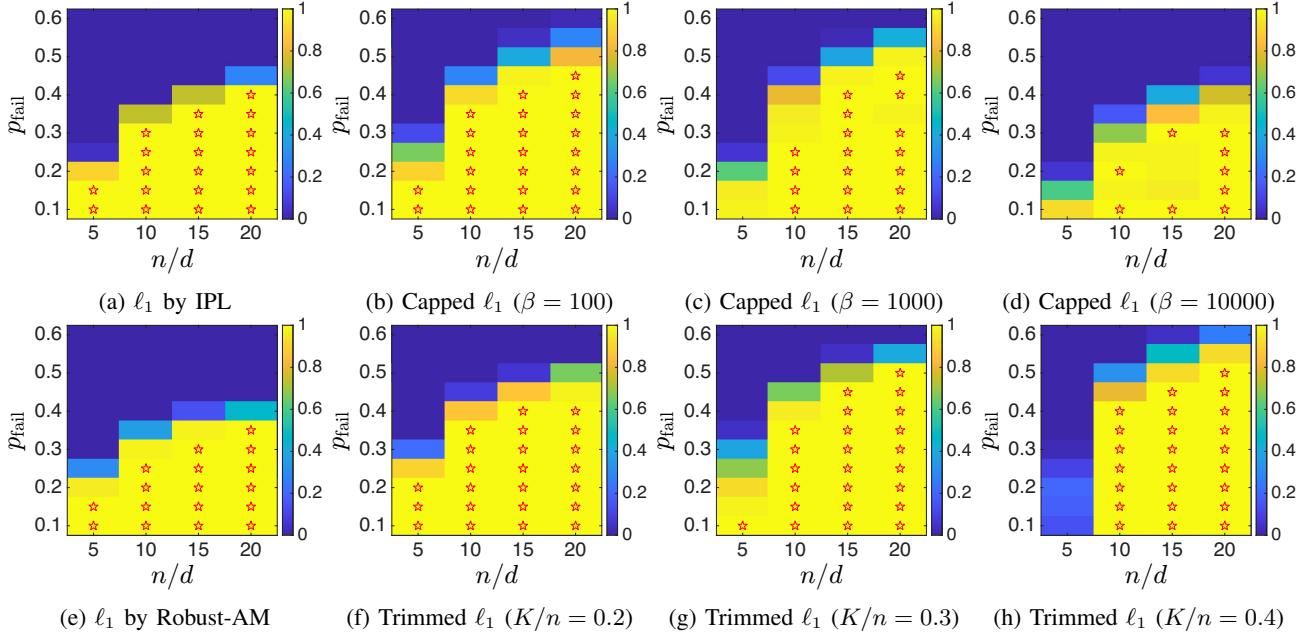


Fig. 1: Success rates of IPL (a), Robust-AM (e), the proposed method with the capped ℓ_1 norm (b)-(d) and the trimmed ℓ_1 norm (f)-(h). The color of each pixel represents success rate, and red pentagrams stand for success rate of 1. The experiment was conducted with $d = 100$, $s = 1$, and ξ_i drawn from the Cauchy distribution.

TABLE II: The running CPU time in seconds (outside the parentheses) and the number of iterations (inside the parentheses) for the case $(d, p_{\text{fail}}, s) = (500, 0.35, 1)$.

n/d	5	10	15	20
ℓ_1 by IPL	30.00 (250.32)	23.78 (172.02)	7.61 (8.86)	9.11 (6.28)
Capped ℓ_1 ($\beta = 100$)	0.84 (359.96)	1.60 (362.76)	1.29 (210.36)	1.23 (161.14)
Capped ℓ_1 ($\beta = 1000$)	0.33 (135.76)	0.48 (101.12)	0.52 (77.48)	0.51 (59.38)
Capped ℓ_1 ($\beta = 10000$)	0.28 (112.10)	0.38 (78.04)	0.37 (50.84)	0.44 (49.82)
Trimmed ℓ_1 ($K/n = 0.2$)	0.47 (137.90)	0.59 (88.80)	0.66 (66.10)	0.77 (62.56)
Trimmed ℓ_1 ($K/n = 0.3$)	0.92 (290.06)	3.49 (619.48)	3.84 (472.74)	3.55 (347.74)
Trimmed ℓ_1 ($K/n = 0.4$)	1.97 (664.14)	2.46 (56.64)	1.83 (44.92)	5.36 (46.82)

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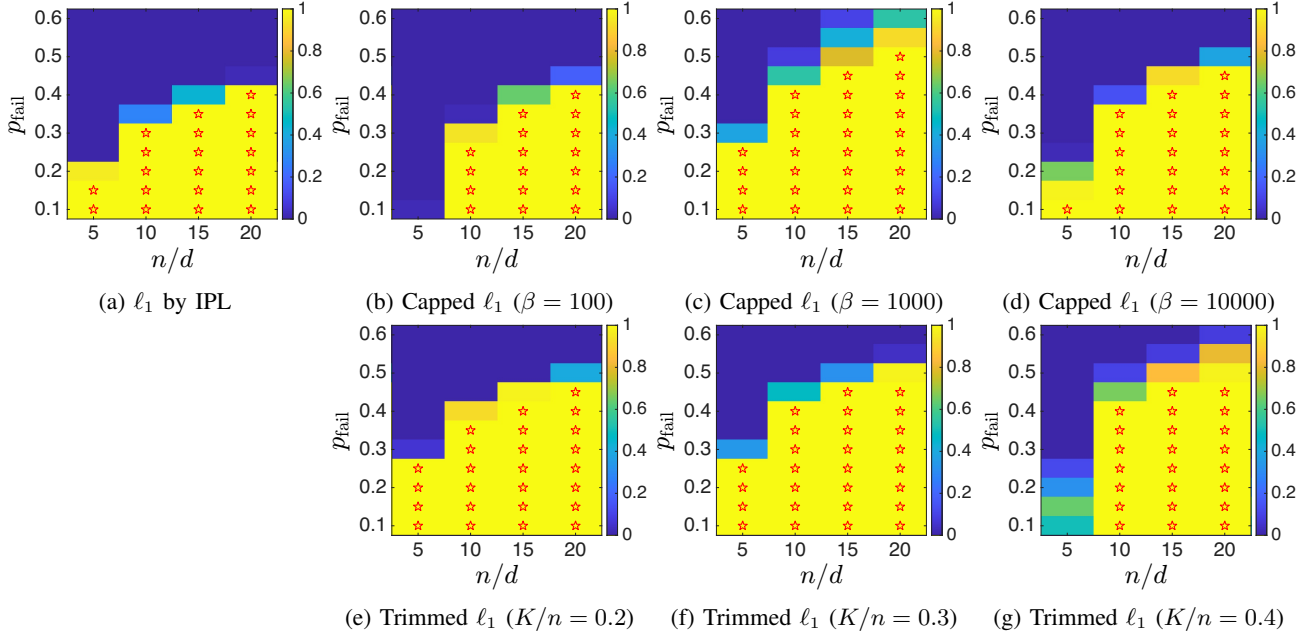


Fig. 2: Same as Fig. 1, except that $d = 500$. We omitted the result of Robust-AM because the subproblem solver (ADMM-LAD) did not reach its stopping criterion within 30 seconds even when $p_{\text{fail}} = 0.1$.

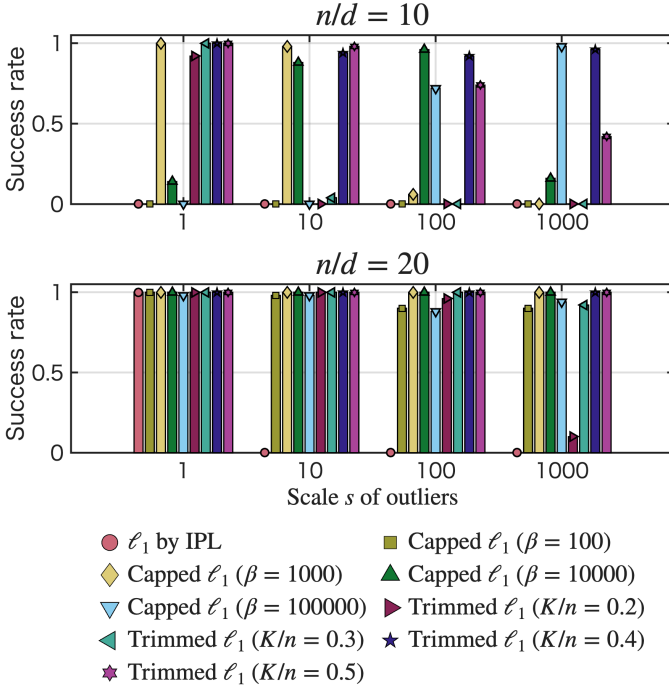


Fig. 3: Success rate for different values of s . (d, p_{fail}) were fixed at (500, 0.4).

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APPENDIX A PROOF OF DEFINITION II.4

We show that a local minimizer $\mathbf{x}^*$ of F is a DC composite critical point of F .

Proof of Definition II.4. It suffices to prove the **Claim 1**: $\partial_L(g \circ \mathfrak{S})(\mathbf{x}^*) \neq \emptyset$ and the **Claim 2**: $\partial_L(f \circ \mathfrak{S})(\mathbf{x}^*) \supset \partial_L(g \circ \mathfrak{S})(\mathbf{x}^*)$ because these two claims imply that $\partial_L(f \circ \mathfrak{S})(\mathbf{x}^*) - \partial_L(g \circ \mathfrak{S})(\mathbf{x}^*) \ni \mathbf{v} - \mathbf{v} = \mathbf{0}$ with some $\mathbf{v} \in \partial_L(g \circ \mathfrak{S})(\mathbf{x}^*)$.

Claim 1: Since $\mathfrak{S}$ is continuously differentiable, $\mathfrak{S}$ is locally Lipschitz continuous (strictly continuous) [43, Thm. 9.7], i.e., for any $\mathbf{x} \in \mathbb{R}^d$, there exists an open neighborhood $\mathcal{N}(\mathbf{x}) \subset \mathbb{R}^d$ of $\mathbf{x}$ such that $\mathfrak{S}$ is Lipschitz continuous on $\mathcal{N}(\mathbf{x})$. Because compositions of locally Lipschitz continuous mappings are locally Lipschitz continuous [43, Exe. 9.8 (c)], $g \circ \mathfrak{S}$ is locally Lipschitz continuous. Thus, we obtain $\partial_L(g \circ \mathfrak{S})(\mathbf{x}^*) \neq \emptyset$ by [43, Thm. 9.13].

Claim 2: By applying the sum rule of Fréchet (regular) subdifferential [43, Cor. 10.9] to $f \circ \mathfrak{S} = F + g \circ \mathfrak{S}$, and using Fermat’s rule $\partial_F F(\mathbf{x}^*) \ni \mathbf{0}$ [43, Thm. 10.1], we have

$$\begin{aligned} \partial_F(f \circ \mathfrak{S})(\mathbf{x}^*) &\supset \partial_F F(\mathbf{x}^*) + \partial_F(g \circ \mathfrak{S})(\mathbf{x}^*) \\ &\supset \partial_F(g \circ \mathfrak{S})(\mathbf{x}^*). \end{aligned}$$

Then, Definition II.2 yields $\partial_L(f \circ \mathfrak{S})(\mathbf{x}^*) \supset \partial_L(g \circ \mathfrak{S})(\mathbf{x}^*)$. $\square$

APPENDIX B PROOF OF DEFINITION III.1

The following facts are required for the proof of Definition III.1.

Fact B.1 (Sum rule of outer limit). For a point sequence $(\mathbf{p}_k)_{k=1}^\infty \subset \mathbb{R}^d$ and a bounded point sequence $(\mathbf{q}_k)_{k=1}^\infty \subset \mathbb{R}^d$, We have $\text{Limsup}_{k \rightarrow \infty}(\mathbf{p}_k + \mathbf{q}_k) \subset \text{Limsup}_{k \rightarrow \infty} \mathbf{p}_k + \text{Limsup}_{k \rightarrow \infty} \mathbf{q}_k$.⁷

Fact B.2 (Boundedness of gradient sequences [51, Fact 2.4 (c)]). In the setting of Definition III.1, $(\nabla^{(\mu_k} f \circ \mathfrak{S})(\mathbf{x}_k))_{k=1}^\infty$ and $(\nabla^{(\mu_k} g \circ \mathfrak{S})(\mathbf{x}_k))_{k=1}^\infty$ are bounded sequences.

Fact B.3 (Gradient sub-consistency [45, Thm. 4.4 (a)]). In the setting of Definition III.1, we have

$$\begin{aligned} \text{Limsup}_{k \rightarrow \infty} \nabla^{(\mu_k} f \circ \mathfrak{S})(\mathbf{x}_k) &\subset \partial_L(f \circ \mathfrak{S})(\bar{\mathbf{x}}), \\ \text{Limsup}_{k \rightarrow \infty} \nabla^{(\mu_k} g \circ \mathfrak{S})(\mathbf{x}_k) &\subset \partial_L(g \circ \mathfrak{S})(\bar{\mathbf{x}}). \end{aligned}$$

Proof of Definition III.1. (a) We can see from Definition B.2 that $(-\nabla^{(\mu_k} g \circ \mathfrak{S})(\mathbf{x}_k))_{k=1}^\infty$ is bounded. Thus, we can employ Definition B.1 with $\mathbf{p}_k := \nabla^{(\mu_k} f \circ \mathfrak{S})(\mathbf{x}_k)$, $\mathbf{q}_k := -\nabla^{(\mu_k} g \circ \mathfrak{S})(\mathbf{x}_k)$ to deduce that

$$\begin{aligned} \text{Limsup}_{k \rightarrow \infty} \nabla F_k(\mathbf{x}_k) &= \text{Limsup}_{k \rightarrow \infty} \nabla^{(\mu_k} f \circ \mathfrak{S} - \mu_k g \circ \mathfrak{S})(\mathbf{x}_k) \\ &\subset \text{Limsup}_{k \rightarrow \infty} \nabla^{(\mu_k} f \circ \mathfrak{S})(\mathbf{x}_k) + \text{Limsup}_{k \rightarrow \infty} (-\nabla^{(\mu_k} g \circ \mathfrak{S})(\mathbf{x}_k)) \\ &= \text{Limsup}_{k \rightarrow \infty} \nabla^{(\mu_k} f \circ \mathfrak{S})(\mathbf{x}_k) - \text{Limsup}_{k \rightarrow \infty} \nabla^{(\mu_k} g \circ \mathfrak{S})(\mathbf{x}_k). \end{aligned}$$

By combining Definition B.3, we get the desired inclusion:

$$\begin{aligned} &\text{Limsup}_{k \rightarrow \infty} \nabla^{(\mu_k} f \circ \mathfrak{S})(\mathbf{x}_k) - \text{Limsup}_{k \rightarrow \infty} \nabla^{(\mu_k} g \circ \mathfrak{S})(\mathbf{x}_k) \\ &\subset \partial_L(f \circ \mathfrak{S})(\bar{\mathbf{x}}) - \partial_L(g \circ \mathfrak{S})(\bar{\mathbf{x}}). \end{aligned}$$

(b) The claim can be derived as follows:

$$\begin{aligned} \liminf_{k \rightarrow \infty} \|\nabla F_k(\mathbf{x}_k)\| &= \liminf_{k \rightarrow \infty} \text{dist}(\mathbf{0}, \nabla F_k(\mathbf{x}_k)) \\ &= \text{dist}\left(\mathbf{0}, \text{Limsup}_{k \rightarrow \infty} \nabla F_k(\mathbf{x}_k)\right) \\ &\geq \text{dist}(\mathbf{0}, \partial_L(f \circ \mathfrak{S})(\bar{\mathbf{x}}) - \partial_L(g \circ \mathfrak{S})(\bar{\mathbf{x}})), \end{aligned}$$

where we used [43, Exe. 4.8] in the second equality and Definition III.1 (a) in the inequality, respectively. $\square$

APPENDIX C PROOF OF DEFINITION III.3

We start with (b) and (c), as their proof are relatively simple.

Proof of Definition III.3 (b) and (c). (b) By applying [45, Prop. 4.2], for any $\mu \in (0, (2\eta)^{-1}]$, we can show that $\nabla^{(\mu} f \circ \mathfrak{S})$ and $\nabla^{(\mu} g \circ \mathfrak{S})$ are Lipschitz continuous, where their Lipschitz constants are $L_{D\mathfrak{S}}L_f + L_{\mathfrak{S}}^2\mu^{-1}$ and $L_{D\mathfrak{S}}L_g + L_{\mathfrak{S}}^2\mu^{-1}$, respectively. Then, $\nabla F^{(\mu)}$ is κ_μ -Lipschitz continuous with $\kappa_\mu := L_{D\mathfrak{S}}(L_f + L_g) + 2L_{\mathfrak{S}}^2\mu^{-1}$. Thus, (12) immediately follows from the descent lemma (see, e.g., [32, Lem. 5.7]).

⁷Although this fact is elementary, we provide a simple proof for our self-containedness. For any $\mathbf{v} \in \text{Limsup}_{k \rightarrow \infty}(\mathbf{p}_k + \mathbf{q}_k)$, there exists a subsequence $(\mathbf{p}_{m(l)} + \mathbf{q}_{m(l)})_{l=1}^\infty \rightarrow \mathbf{v}$, where $m: \mathbb{N} \rightarrow \mathbb{N}$ is monotonically increasing. From the boundedness of $(\mathbf{q}_k)_{k=1}^\infty$ and the Bolzano-Weierstrass theorem (see, e.g., [50, Thm. 3.24]), we get $\mathbf{q}_{m(l)} \rightarrow \exists \mathbf{q} \in \mathbb{R}^d$ by passing through further a subsequence of $(\mathbf{q}_{m(l)})_{l=1}^\infty$ (and renaming it as $(\mathbf{q}_{m(l)})_{l=1}^\infty$). Hence, we have $\mathbf{q} \in \text{Limsup}_{k \rightarrow \infty} \mathbf{q}_k$ and $\mathbf{v} - \mathbf{q} = \lim_{l \rightarrow \infty} (\mathbf{p}_{m(l)} + \mathbf{q}_{m(l)}) - \mathbf{q}_{m(l)} = \lim_{l \rightarrow \infty} \mathbf{p}_{m(l)} \in \text{Limsup}_{k \rightarrow \infty} \mathbf{p}_k$, which implies $\mathbf{v} = (\mathbf{v} - \mathbf{q}) + \mathbf{q} \in \text{Limsup}_{k \rightarrow \infty} \mathbf{p}_k + \text{Limsup}_{k \rightarrow \infty} \mathbf{q}_k$.

(c) For any $\mu \in (0, (2\eta)^{-1}]$ and $\mathbf{x} \in \mathbb{R}^d$, we can deduce from the definition of the Moreau envelope that

$$\begin{aligned} (\mu \hat{f} \circ \hat{\mathfrak{S}})(\mathbf{x}) &= \min_{\mathbf{z} \in \mathbb{R}^n, t \in \mathbb{R}} \left\{ f(\mathbf{z}) + t + \frac{1}{2\mu} \left\| \begin{bmatrix} \mathfrak{S}(\mathbf{x}) \\ h(\mathbf{x}) \end{bmatrix} - \begin{bmatrix} \mathbf{z} \\ t \end{bmatrix} \right\|^2 \right\} \\ &= (\mu f \circ \mathfrak{S})(\mathbf{x}) + \min_{t \in \mathbb{R}} \left\{ t + \frac{1}{2\mu} |t - h(\mathbf{x})|^2 \right\} \\ &= (\mu f \circ \mathfrak{S})(\mathbf{x}) + h(\mathbf{x}) - \frac{\mu}{2}, \\ (\mu \hat{g} \circ \hat{\mathfrak{S}})(\mathbf{x}) &= (\mu g \circ \mathfrak{S})(\mathbf{x}). \end{aligned}$$

Then, we get $\hat{F}^{(\mu)}(\mathbf{x}) = h(\mathbf{x}) - \frac{\mu}{2} + F^{(\mu)}(\mathbf{x})$. On the other hand, the descent lemma for h implies that, for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$,

$$h(\mathbf{y}) - \frac{\mu}{2} \leq h(\mathbf{x}) - \frac{\mu}{2} + \langle \nabla h(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \frac{L_{\nabla h}}{2} \|\mathbf{y} - \mathbf{x}\|^2 \quad (20)$$

holds. Therefore, by summing (12) and (20), we obtain

$$\hat{F}^{(\mu)}(\mathbf{y}) \leq \hat{F}^{(\mu)}(\mathbf{x}) + \langle \nabla \hat{F}^{(\mu)}, \mathbf{y} - \mathbf{x} \rangle + \frac{\hat{\kappa}_\mu}{2} \|\mathbf{y} - \mathbf{x}\|^2,$$

where $\hat{\kappa}_\mu := \kappa_\mu + L_{\nabla h}$. $\square$

We use the following fact and lemma to show Definition III.3 (a).

Fact C.1 (Weak convexity of composite functions [52, Lem. 4.2]). Let a function $\psi : \mathbb{R}^n \rightarrow \mathbb{R}$ be convex and L_ψ -Lipschitz continuous, and suppose that a mapping $\mathcal{F} : \mathbb{R}^d \rightarrow \mathbb{R}^n$ is differentiable and $D\mathcal{F}$ is $L_{D\mathcal{F}}$ -Lipschitz continuous. Then, $\psi \circ \mathcal{F}$ is $L_\psi L_{D\mathcal{F}}$ -weakly convex.

Lemma C.2. Let $\mathcal{X}, \mathcal{Z}$ be Euclidean spaces. Consider a closed set $\mathcal{D} \subset \mathcal{Z}$ and its complement $\mathcal{E} := \mathcal{Z} \setminus \mathcal{D}$. For a function $J : \mathcal{Z} \rightarrow \mathbb{R}$ and a mapping $\mathcal{F} : \mathcal{X} \rightarrow \mathcal{Z}$, assume that

- (i) J is differentiable and ∇J is $L_{\nabla J}$ -Lipschitz continuous on $\mathcal{Z}$.
- (ii) J is L_J -Lipschitz continuous on $\mathcal{Z}$, and thus, $\|\nabla J(\cdot)\|$ is bounded above by L_J on $\mathcal{Z}$ [43, Thm. 9.7], i.e.,
$$(\forall \mathbf{z} \in \mathcal{Z}) \quad \|\nabla J(\mathbf{z})\| \leq L_J.$$
- (iii) J is affine on $\mathcal{D}$, i.e.,
$$(\exists \mathbf{u} \in \mathcal{Z}, \exists \tau \in \mathbb{R}, \forall \mathbf{z} \in \mathcal{Z}) \quad J(\mathbf{z}) = \langle \mathbf{u}, \mathbf{z} \rangle + \tau.$$
- (iv) $\mathcal{F}$ is differentiable and $D\mathcal{F}$ is $L_{D\mathcal{F}}$ -Lipschitz continuous on $\mathcal{X}$.
- (v) $\|D\mathcal{F}(\cdot)\|_{\text{op}}$ is bounded above by $M > 0$ on $\mathcal{F}^{-1}(\mathcal{E}) := \{\mathbf{v} \in \mathcal{X} \mid \mathcal{F}(\mathbf{v}) \in \mathcal{E}\}$.

Then, $\nabla(J \circ \mathcal{F})$ is $L_{\nabla(J \circ \mathcal{F})}$ -Lipschitz continuous on $\mathcal{X}$ with $L_{\nabla(J \circ \mathcal{F})} := M^2 L_{\nabla J} + L_J L_{D\mathcal{F}}$.

Proof. By using the chain rule for $\nabla(J \circ \mathcal{F})$, it holds for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$ that

$$\begin{aligned} &\|\nabla(J \circ \mathcal{F})(\mathbf{x}) - \nabla(J \circ \mathcal{F})(\mathbf{y})\| \\ &= \| (D\mathcal{F}(\mathbf{x}))^* [\nabla J(\mathcal{F}(\mathbf{x}))] - (D\mathcal{F}(\mathbf{y}))^* [\nabla J(\mathcal{F}(\mathbf{y}))] \| \\ &\leq \| (D\mathcal{F}(\mathbf{x}))^* [\nabla J(\mathcal{F}(\mathbf{x})) - \nabla J(\mathcal{F}(\mathbf{y}))] \| \\ &\quad + \| (D\mathcal{F}(\mathbf{x}) - D\mathcal{F}(\mathbf{y}))^* [\nabla J(\mathcal{F}(\mathbf{y}))] \| \\ &\leq \|D\mathcal{F}(\mathbf{x})\|_{\text{op}} \|\nabla J(\mathcal{F}(\mathbf{x})) - \nabla J(\mathcal{F}(\mathbf{y}))\| \\ &\quad + \|D\mathcal{F}(\mathbf{x}) - D\mathcal{F}(\mathbf{y})\|_{\text{op}} \|\nabla J(\mathcal{F}(\mathbf{y}))\| \\ &\leq \|D\mathcal{F}(\mathbf{x})\|_{\text{op}} \|\nabla J(\mathcal{F}(\mathbf{x})) - \nabla J(\mathcal{F}(\mathbf{y}))\| + L_J L_{D\mathcal{F}} \|\mathbf{x} - \mathbf{y}\|, \end{aligned} \quad (21)$$

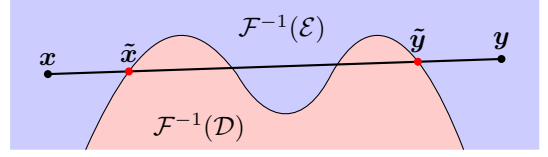


Fig. 4: Illustration of the positions of $\tilde{\mathbf{x}}$ and $\tilde{\mathbf{y}}$

where the last inequality follows from the assumptions (ii) and (iv). (Note: X^* stands for the *adjoint operator* of the linear operator X . When X is regarded as a matrix, X^* coincides with its transpose X^T .)

Here, we consider two cases according to whether the open line segment $\ell(\mathbf{x}, \mathbf{y}) := \{t\mathbf{x} + (1-t)\mathbf{y} \mid 0 < t < 1\}$ intersects $\mathcal{F}^{-1}(\mathcal{D})$ or not.

Case 1: $\ell(\mathbf{x}, \mathbf{y}) \cap \mathcal{F}^{-1}(\mathcal{D}) = \emptyset$

Since $\|\nabla(J \circ \mathcal{F})(\mathbf{x}) - \nabla(J \circ \mathcal{F})(\mathbf{y})\| \leq L_{\nabla(J \circ \mathcal{F})} \|\mathbf{x} - \mathbf{y}\|$ obviously holds in the trivial case $\mathbf{x} = \mathbf{y}$, we assume $\mathbf{x} \neq \mathbf{y}$. From $\ell(\mathbf{x}, \mathbf{y}) \cap \mathcal{F}^{-1}(\mathcal{D}) = \emptyset$, we have

$$\ell(\mathbf{x}, \mathbf{y}) \subset \mathcal{X} \setminus \mathcal{F}^{-1}(\mathcal{D}) = \mathcal{F}^{-1}(\mathcal{Z} \setminus \mathcal{D}) = \mathcal{F}^{-1}(\mathcal{E}).$$

Then, the assumption (v) implies that

$$\|D\mathcal{F}(\mathbf{v})\|_{\text{op}} \leq M \quad (\mathbf{v} \in \ell(\mathbf{x}, \mathbf{y})).$$

Moreover, from the continuity of $\|D\mathcal{F}(\cdot)\|_{\text{op}}$, we have

$$\|D\mathcal{F}(\mathbf{x})\|_{\text{op}} = \lim_{t \nearrow 1} \|D\mathcal{F}(t\mathbf{x} + (1-t)\mathbf{y})\|_{\text{op}} \leq \lim_{t \nearrow 1} M = M, \quad (22)$$

and we also get $\|D\mathcal{F}(\mathbf{y})\|_{\text{op}} \leq M$ in the same manner. Thus, the mean value inequality (see, e.g., [53, p.155]) yields that

$$\begin{aligned} \|\mathcal{F}(\mathbf{x}) - \mathcal{F}(\mathbf{y})\| &\leq \sup_{0 \leq t \leq 1} \|D\mathcal{F}(t\mathbf{x} + (1-t)\mathbf{y})\|_{\text{op}} \|\mathbf{x} - \mathbf{y}\| \\ &= \sup_{\mathbf{v} \in \ell(\mathbf{x}, \mathbf{y}) \cup \{\mathbf{x}, \mathbf{y}\}} \|D\mathcal{F}(\mathbf{v})\|_{\text{op}} \|\mathbf{x} - \mathbf{y}\| \leq M \|\mathbf{x} - \mathbf{y}\|. \end{aligned}$$

Therefore, by combining the assumption (i) and (22), we have

$$\begin{aligned} &\|D\mathcal{F}(\mathbf{x})\|_{\text{op}} \|\nabla J(\mathcal{F}(\mathbf{x})) - \nabla J(\mathcal{F}(\mathbf{y}))\| \\ &\leq \|D\mathcal{F}(\mathbf{x})\|_{\text{op}} M L_{\nabla J} \|\mathbf{x} - \mathbf{y}\| \leq M^2 L_{\nabla J} \|\mathbf{x} - \mathbf{y}\|. \end{aligned} \quad (23)$$

From (21) and (23), we obtain

$$\|\nabla(J \circ \mathcal{F})(\mathbf{x}) - \nabla(J \circ \mathcal{F})(\mathbf{y})\| \leq L_{\nabla(J \circ \mathcal{F})} \|\mathbf{x} - \mathbf{y}\|.$$

Case 2: $\ell(\mathbf{x}, \mathbf{y}) \cap \mathcal{F}^{-1}(\mathcal{D}) \neq \emptyset$

We define points $\tilde{\mathbf{x}}, \tilde{\mathbf{y}}$ on the closed line segment $\text{cl}(\ell(\mathbf{x}, \mathbf{y})) = \ell(\mathbf{x}, \mathbf{y}) \cup \{\mathbf{x}, \mathbf{y}\}$ as

$$\tilde{\mathbf{x}} := \underset{\mathbf{v} \in \mathcal{J}}{\text{argmin}} \|\mathbf{x} - \mathbf{v}\|, \quad \tilde{\mathbf{y}} := \underset{\mathbf{v} \in \mathcal{J}}{\text{argmin}} \|\mathbf{y} - \mathbf{v}\|$$

$$\text{with } \mathcal{J} := \text{cl}(\ell(\mathbf{x}, \mathbf{y})) \cap \mathcal{F}^{-1}(\mathcal{D})$$

(see Fig. 4 for an intuitive illustration). Note that $\tilde{\mathbf{x}}$ and $\tilde{\mathbf{y}}$ are well-defined due to the compactness of $\mathcal{J}$. For $\tilde{\mathbf{x}}, \tilde{\mathbf{y}}$, there exist $t_{\tilde{\mathbf{x}}}, t_{\tilde{\mathbf{y}}} \in [0, 1]$ such that $t_{\tilde{\mathbf{y}}} \leq t_{\tilde{\mathbf{x}}}$ and $\tilde{\mathbf{x}} = t_{\tilde{\mathbf{x}}} \mathbf{x} + (1 - t_{\tilde{\mathbf{x}}}) \mathbf{y}$, $\tilde{\mathbf{y}} = t_{\tilde{\mathbf{y}}} \mathbf{x} + (1 - t_{\tilde{\mathbf{y}}}) \mathbf{y}$. From $\mathbf{x} - \tilde{\mathbf{x}} = (1 - t_{\tilde{\mathbf{x}}})(\mathbf{x} - \mathbf{y})$, $\tilde{\mathbf{x}} - \tilde{\mathbf{y}} = (t_{\tilde{\mathbf{x}}} - t_{\tilde{\mathbf{y}}})(\mathbf{x} - \mathbf{y})$, $\tilde{\mathbf{y}} - \mathbf{y} = t_{\tilde{\mathbf{y}}}(\mathbf{x} - \mathbf{y})$, we have

$$\|\mathbf{x} - \mathbf{y}\| = \|\mathbf{x} - \tilde{\mathbf{x}}\| + \|\tilde{\mathbf{x}} - \tilde{\mathbf{y}}\| + \|\tilde{\mathbf{y}} - \mathbf{y}\|. \quad (24)$$

In what follows, we consider the three line segments $\ell(\mathbf{x}, \tilde{\mathbf{x}})$, $\ell(\tilde{\mathbf{y}}, \mathbf{y})$, and $\ell(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$.

Because $\ell(\mathbf{x}, \tilde{\mathbf{x}}) \cap \mathcal{F}^{-1}(\mathcal{D}) = \emptyset$ and $\ell(\tilde{\mathbf{y}}, \mathbf{y}) \cap \mathcal{F}^{-1}(\mathcal{D}) = \emptyset$ follow from the definition of $\tilde{\mathbf{x}}$ and $\tilde{\mathbf{y}}$, the same argument as the Case 1 yields that

$$\|\nabla(J \circ \mathcal{F})(\mathbf{x}) - \nabla(J \circ \mathcal{F})(\tilde{\mathbf{x}})\| \leq L_{\nabla(J \circ \mathcal{F})} \|\mathbf{x} - \tilde{\mathbf{x}}\|, \quad (25)$$

$$\|\nabla(J \circ \mathcal{F})(\mathbf{y}) - \nabla(J \circ \mathcal{F})(\tilde{\mathbf{y}})\| \leq L_{\nabla(J \circ \mathcal{F})} \|\mathbf{y} - \tilde{\mathbf{y}}\|. \quad (26)$$

For $\ell(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$, the assumption (iii) together with $\tilde{\mathbf{x}}, \tilde{\mathbf{y}} \in \mathcal{F}^{-1}(\mathcal{D})$ implies that

$$\|\nabla J(\mathcal{F}(\tilde{\mathbf{x}})) - \nabla J(\mathcal{F}(\tilde{\mathbf{y}}))\| = \|\mathbf{u} - \mathbf{u}\| = 0,$$

and then, (21) with the substitution $(\mathbf{x}, \mathbf{y}) = (\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$ yields

$$\|\nabla(J \circ \mathcal{F})(\tilde{\mathbf{x}}) - \nabla(J \circ \mathcal{F})(\tilde{\mathbf{y}})\| \leq L_J L_{\mathcal{D}\mathcal{F}} \|\tilde{\mathbf{x}} - \tilde{\mathbf{y}}\|. \quad (27)$$

By using (25) to (27), we obtain

$$\begin{aligned} & \|\nabla(J \circ \mathcal{F})(\mathbf{x}) - \nabla(J \circ \mathcal{F})(\mathbf{y})\| \\ & \leq \|\nabla(J \circ \mathcal{F})(\mathbf{x}) - \nabla(J \circ \mathcal{F})(\tilde{\mathbf{x}})\| \\ & \quad + \|\nabla(J \circ \mathcal{F})(\tilde{\mathbf{x}}) - \nabla(J \circ \mathcal{F})(\tilde{\mathbf{y}})\| \\ & \quad + \|\nabla(J \circ \mathcal{F})(\tilde{\mathbf{y}}) - \nabla(J \circ \mathcal{F})(\mathbf{y})\| \\ & \leq L_{\nabla(J \circ \mathcal{F})} \|\mathbf{x} - \tilde{\mathbf{x}}\| + L_J L_{\mathcal{D}\mathcal{F}} \|\tilde{\mathbf{x}} - \tilde{\mathbf{y}}\| + L_{\nabla(J \circ \mathcal{F})} \|\tilde{\mathbf{y}} - \mathbf{y}\| \\ & \leq L_{\nabla(J \circ \mathcal{F})} (\|\mathbf{x} - \tilde{\mathbf{x}}\| + \|\tilde{\mathbf{x}} - \tilde{\mathbf{y}}\| + \|\tilde{\mathbf{y}} - \mathbf{y}\|) \\ & = L_{\nabla(J \circ \mathcal{F})} \|\mathbf{x} - \mathbf{y}\|. \quad (\because (24)) \end{aligned}$$

□

Proof of Definition III.3 (a). The proof is completed by showing that there exist $\varpi'_1, \varpi''_1, \varpi_2, \in \mathbb{R}_{++}$ such that

$$\begin{aligned} & -(\mu g \circ \mathfrak{S}_{\text{RPR}})(\mathbf{y}) \leq -(\mu g \circ \mathfrak{S}_{\text{RPR}})(\mathbf{x}) \\ & \quad - \langle \nabla(\mu g \circ \mathfrak{S}_{\text{RPR}})(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \frac{\varpi'_1}{2} \|\mathbf{y} - \mathbf{x}\|^2, \quad (28) \\ & (\mu f \circ \mathfrak{S}_{\text{RPR}})(\mathbf{y}) \leq (\mu f \circ \mathfrak{S}_{\text{RPR}})(\mathbf{x}) \\ & \quad + \langle \nabla(\mu f \circ \mathfrak{S}_{\text{RPR}})(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \frac{\varpi''_1 + \varpi_2 \mu^{-1}}{2} \|\mathbf{y} - \mathbf{x}\|^2, \quad (29) \end{aligned}$$

because (12) is obtained by summing (28) and (29), and by setting $\varpi_1 := \varpi'_1 + \varpi''_1$. In the following, we show (28) and (29) separately.

For (28), it is enough to prove the following inequality:

$$\begin{aligned} & (\mu g \circ \mathfrak{S}_{\text{RPR}})(\mathbf{y}) + \frac{\varpi'_1}{2} \|\mathbf{y}\|^2 \geq (\mu g \circ \mathfrak{S}_{\text{RPR}})(\mathbf{x}) + \frac{\varpi'_1}{2} \|\mathbf{x}\|^2 \\ & \quad + \langle \nabla(\mu g \circ \mathfrak{S}_{\text{RPR}})(\mathbf{x}), \mathbf{x} \rangle + \varpi'_1 \mathbf{x}, \mathbf{y} - \mathbf{x}. \end{aligned}$$

This is equivalent to the convexity of $(\mu g \circ \mathfrak{S}_{\text{RPR}}) + \frac{\varpi'_1}{2} \|\cdot\|^2$, that is, ϖ'_1 -weak convexity of $\mu g \circ \mathfrak{S}_{\text{RPR}}$. To show this weak convexity, we can use Definition C.1 with $\psi := \mu g$ and $\mathcal{F} := \mathfrak{S}_{\text{RPR}}$. Indeed, the assumptions of Definition C.1 are easily checked as follows: (i) μg is L_g -Lipschitz continuous because g is L_g -Lipschitz continuous (see [40, Lem. 3.3]); (ii) since every choice of g in Table I is convex, μg is also convex [32, Thm. 6.55]; (iii) the derivative $\mathcal{D}\mathfrak{S}_{\text{RPR}} : \mathbf{x} \mapsto [2\langle \mathbf{a}_1, \mathbf{x} \rangle \mathbf{a}_1, 2\langle \mathbf{a}_2, \mathbf{x} \rangle \mathbf{a}_2, \dots, 2\langle \mathbf{a}_n, \mathbf{x} \rangle \mathbf{a}_n]^T$ is

$(2\sqrt{\sum_{i=1}^n \|\mathbf{a}_i\|^4})$ -Lipschitz continuous because we have, for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$,

$$\begin{aligned} & \|\mathcal{D}\mathfrak{S}_{\text{RPR}}(\mathbf{x}) - \mathcal{D}\mathfrak{S}_{\text{RPR}}(\mathbf{y})\|_{\text{op}} \\ & = \sup_{\|\mathbf{v}\| \leq 1} \|(\mathcal{D}\mathfrak{S}_{\text{RPR}}(\mathbf{x}) - \mathcal{D}\mathfrak{S}_{\text{RPR}}(\mathbf{y}))[\mathbf{v}]\| \\ & = \sup_{\|\mathbf{v}\| \leq 1} \sqrt{\sum_{i=1}^n (2\langle \mathbf{a}_i, \mathbf{x} - \mathbf{y} \rangle \langle \mathbf{a}_i, \mathbf{v} \rangle)^2} \\ & \leq \sup_{\|\mathbf{v}\| \leq 1} \sqrt{\sum_{i=1}^n (2\|\mathbf{a}_i\| \|\mathbf{x} - \mathbf{y}\| \|\mathbf{a}_i\| \|\mathbf{v}\|)^2} \\ & \leq 2 \sqrt{\sum_{i=1}^n \|\mathbf{a}_i\|^4} \|\mathbf{x} - \mathbf{y}\|. \quad (30) \end{aligned}$$

As a result, the weak convexity of $\mu g \circ \mathfrak{S}_{\text{RPR}}$ is proven, and therefore, (28) holds with $\varpi'_1 := 2L_g \sqrt{\sum_{i=1}^n \|\mathbf{a}_i\|^4}$.

To prove (29), we show that $\nabla(\mu f \circ \mathfrak{S}_{\text{RPR}})$ is $(\varpi''_1 + \varpi_2 \mu^{-1})$ -Lipschitz continuous, which is a sufficient condition of (29) as stated in the descent lemma (see, e.g., [32, Lem. 5.7]). We note that all f in Table I have the form $f(\mathbf{z}) = \lambda \sum_{i=1}^n |\mathbf{z}|_i$ ($\mathbf{z} \in \mathbb{R}^n$) with some $\lambda \in \mathbb{R}_{++}$, and thereby, their Moreau envelopes can be derived as $\mu f(\mathbf{z}) = \sum_{i=1}^n \mu(\lambda|\cdot|)([\mathbf{z}]_i) = \sum_{i=1}^n \hat{r}_{\lambda, \mu}([\mathbf{z}]_i)$ [32, Lem. 6.57, Exm. 6.59] (see Table I for the definition of $\hat{r}_{\lambda, \mu}$). In order to invoke Definition C.2 with $J := \hat{r}_{\lambda, \mu}$, $\mathcal{F} := \mathfrak{S}_{\text{RPR}, i} := \langle \mathbf{a}_i, \cdot \rangle^2 - [\mathbf{b}]_i$ ($i \in \{1, 2, \dots, n\}$), and $(\mathcal{D}, \mathcal{E}) := ([\mu\lambda, \infty), (-\infty, \mu\lambda))$, we check that the assumptions (i)-(v) in Definition C.2 hold.

- (i) $\nabla \hat{r}_{\lambda, \mu} = \nabla \mu(\lambda|\cdot|)$ is μ^{-1} -Lipschitz continuous from [32, Thm. 6.60].
- (ii) From the definition of $\hat{r}_{\lambda, \mu}$, it is obviously λ -Lipschitz continuous.
- (iii) $\hat{r}_{\lambda, \mu}(t) = t - \mu\lambda^2/2$ for $t \geq \mu\lambda$, i.e., it is affine on $[\mu\lambda, \infty)$.
- (iv) $\mathcal{D}\mathfrak{S}_{\text{RPR}, i} = 2\langle \mathbf{a}_i, \cdot \rangle \mathbf{a}_i^T$ is $(2\|\mathbf{a}_i\|^2)$ -Lipschitz continuous.
- (v) It holds for any $\mathbf{x} \in \mathfrak{S}_{\text{RPR}, i}^{-1}((-\infty, \mu\lambda))$ that $\|\mathcal{D}\mathfrak{S}_{\text{RPR}, i}(\mathbf{x})\|_{\text{op}} = \|2\langle \mathbf{a}_i, \mathbf{x} \rangle \mathbf{a}_i^T\|_{\text{op}} = 2\|\mathbf{a}_i\| |\langle \mathbf{a}_i, \mathbf{x} \rangle| = 2\|\mathbf{a}_i\| \sqrt{\mathfrak{S}_{\text{RPR}, i}(\mathbf{x}) + [\mathbf{b}]_i} < 2\|\mathbf{a}_i\| \sqrt{\mu\lambda + [\mathbf{b}]_i}$.

From above, Definition C.2 yields that $\nabla(\hat{r}_{\lambda, \mu} \circ \mathfrak{S}_{\text{RPR}, i})$ is Lipschitz continuous with a Lipschitz constant $4\|\mathbf{a}_i\|^2(\mu\lambda + [\mathbf{b}]_i)\mu^{-1} + \lambda(2\|\mathbf{a}_i\|^2) = 6\|\mathbf{a}_i\|^2\lambda + 4\|\mathbf{a}_i\|^2[\mathbf{b}]_i\mu^{-1}$. Then, for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$, we have

$$\begin{aligned} & \|\nabla(\mu f \circ \mathfrak{S}_{\text{RPR}})(\mathbf{x}) - \nabla(\mu f \circ \mathfrak{S}_{\text{RPR}})(\mathbf{y})\|^2 \\ & \leq \sum_{i=1}^n (6\|\mathbf{a}_i\|^2\lambda + 4\|\mathbf{a}_i\|^2[\mathbf{b}]_i\mu^{-1})^2 ([\mathbf{x}]_i - [\mathbf{y}]_i)^2 \\ & \leq \max_{1 \leq i \leq n} (6\|\mathbf{a}_i\|^2\lambda + 4\|\mathbf{a}_i\|^2[\mathbf{b}]_i\mu^{-1})^2 \sum_{j=1}^n ([\mathbf{x}]_j - [\mathbf{y}]_j)^2 \\ & = \left(\max_{1 \leq i \leq n} (6\|\mathbf{a}_i\|^2\lambda + 4\|\mathbf{a}_i\|^2[\mathbf{b}]_i\mu^{-1}) \right)^2 \sum_{j=1}^n ([\mathbf{x}]_j - [\mathbf{y}]_j)^2 \\ & \leq \left(6\lambda \max_{1 \leq i \leq n} \|\mathbf{a}_i\|^2 + 4\mu^{-1} \max_{1 \leq i \leq n} \|\mathbf{a}_i\|^2 [\mathbf{b}]_i \right)^2 \|\mathbf{x} - \mathbf{y}\|^2. \end{aligned}$$

Thus, $\nabla(\mu f \circ \mathfrak{S}_{\text{RPR}})$ is $(\varpi''_1 + \varpi_2 \mu^{-1})$ -Lipschitz continuous, where $\varpi''_1 := 6\lambda \max_{1 \leq i \leq n} \|\mathbf{a}_i\|^2$, $\varpi_2 := 4 \max_{1 \leq i \leq n} \|\mathbf{a}_i\|^2 [\mathbf{b}]_i$.

□

APPENDIX D PROOF OF DEFINITION III.6

We show that the choices of initial guesses $(\gamma_{\text{init},k})_{k=1}^\infty$ in Definition III.6 satisfy Definition III.5.

Proof of Definition III.6. (a) The inequality (15) is obviously satisfied with $\delta := 2(1-c)$.

(b) Since $(\mu_k)_{k=1}^\infty$ is non-increasing from (13)(iii), $\kappa_{\mu_k} := \varpi_1 + \varpi_2/\mu_k$ is non-decreasing, and then, we have $\kappa_{\mu_k} \geq \kappa_{\mu_1}$ ($k \in \mathbb{N}$). Hence, the inequality $\gamma_{\text{init},k} := \gamma_{\text{init}} \geq \delta \kappa_{\mu_k}^{-1}$ holds with $\delta := \gamma_{\text{init}} \kappa_{\mu_1}$.

(c) By the induction, we show $\gamma_{\text{init},k} \geq \delta \kappa_{\mu_k}^{-1}$ ($\forall k \in \mathbb{N}$) in (15) with $\delta := \delta_0 := \min\{\gamma_0 \kappa_{\mu_1}, 2(1-c)\rho\}$. For $\gamma_{\text{init},1} := \gamma_0$, we have $\gamma_{\text{init},1} = (\gamma_0 \kappa_{\mu_1}) \kappa_{\mu_1}^{-1} \geq \delta_0 \kappa_{\mu_1}^{-1}$. On the other hand, by using Definition III.4 (a) and the induction hypothesis $\gamma_{\text{init},k} \geq \delta_0 \kappa_{\mu_k}^{-1}$, we obtain $\gamma_{\text{init},k+1} := \gamma_k \geq \min\{\gamma_{\text{init},k}, 2(1-c)\kappa_{\mu_k}^{-1}\rho\} \geq \min\{\delta_0, 2(1-c)\rho\} \kappa_{\mu_k}^{-1} = \delta_0 \kappa_{\mu_k}^{-1} \geq \delta_0 \kappa_{\mu_{k+1}}^{-1}$, where the last equality follows from $2(1-c)\rho \geq \delta_0$, and the last inequality holds since $(\kappa_{\mu_k})_{k=1}^\infty$ is non-decreasing. This completes the inductive proof. $\square$

APPENDIX E PROOF OF DEFINITION III.7

We make use of the next fact to prove Definition III.7.

Fact E.1 (Properties of Moreau envelope of Lipschitz continuous function). Let a function $\psi : \mathbb{R}^n \rightarrow \mathbb{R}$ be η_ψ -weakly convex and L_ψ -Lipschitz continuous. Then, for any $\mathbf{z} \in \mathbb{R}^n$ and $\mu_I, \mu_{II} \in (0, \eta_\psi)$ such that $\mu_{II} < \mu_I$, we have

$$\mu_I \psi(\mathbf{z}) \leq \mu_{II} \psi(\mathbf{z}) \leq \mu_I \psi(\mathbf{z}) + (\mu_I - \mu_{II}) L_\psi^2 \quad (31)$$

(see [45, Eq. (21)] and the subsequent discussion in [45]). Moreover, by letting $\mu_{II} \searrow 0$, we obtain the following (see Definition II.6 (a)):

$$\mu_I \psi(\mathbf{z}) \leq \psi(\mathbf{z}) \leq \mu_I \psi(\mathbf{z}) + \mu_I L_\psi^2. \quad (32)$$

Proof of Definition III.7. This proof is inspired by those for [40, Thm. 4.1] and [45, Thm. 4.8].

(a) Since the stepsize γ_k output by Algorithm 2 satisfies the Armijo condition (14), we have

$$F_k(\mathbf{x}_{k+1}) \leq F_k(\mathbf{x}_k) - c\gamma_k \|\nabla F_k(\mathbf{x}_k)\|^2 \quad (k \in \mathbb{N}). \quad (33)$$

On the other hand, by virtue of (31) and $\mu_{k+1} \leq \mu_k$ (see (13)(iii)), the following inequality holds for any $\mathbf{x} \in \mathbb{R}^d$:

$$\begin{aligned} F_k(\mathbf{x}) &= \mu_k f(\mathfrak{S}(\mathbf{x})) - \mu_k g(\mathfrak{S}(\mathbf{x})) \\ &\geq \mu_k f(\mathfrak{S}(\mathbf{x})) - \mu_{k+1} g(\mathfrak{S}(\mathbf{x})) \\ &\geq \mu_{k+1} f(\mathfrak{S}(\mathbf{x})) - (\mu_k - \mu_{k+1}) L_f^2 - \mu_{k+1} g(\mathfrak{S}(\mathbf{x})) \\ &= F_{k+1}(\mathbf{x}) - (\mu_k - \mu_{k+1}) L_f^2. \end{aligned} \quad (34)$$

By combining (34) with $\mathbf{x} := \mathbf{x}_{k+1}$ and (33), we obtain

$$F_{k+1}(\mathbf{x}_{k+1}) \leq F_k(\mathbf{x}_k) - c\gamma_k \|\nabla F_k(\mathbf{x}_k)\|^2 + (\mu_k - \mu_{k+1}) L_f^2, \quad (35)$$

and thus,

$$F_{k+1}(\mathbf{x}_{k+1}) \leq F_k(\mathbf{x}_k) + (\mu_k - \mu_{k+1}) L_f^2. \quad (36)$$

By summing up (35) from $k = \underline{k}$ to $\bar{k}$, we deduce that

$$\sum_{k=\underline{k}}^{\bar{k}} c\gamma_k \|\nabla F_k(\mathbf{x}_k)\|^2 \leq F_{\underline{k}}(\mathbf{x}_{\underline{k}}) - F_{\bar{k}+1}(\mathbf{x}_{\bar{k}+1}) + (\mu_{\underline{k}} - \mu_{\bar{k}+1}) L_f^2.$$

We use (36) repeatedly to get

$$\sum_{k=\underline{k}}^{\bar{k}} c\gamma_k \|\nabla F_k(\mathbf{x}_k)\|^2 \leq F_1(\mathbf{x}_1) - F_{\bar{k}+1}(\mathbf{x}_{\bar{k}+1}) + (\mu_1 - \mu_{\bar{k}+1}) L_f^2.$$

Here, we see from (32) in Definition E.1 that

$$\begin{aligned} F_{\bar{k}+1}(\mathbf{x}_{\bar{k}+1}) &= \mu_{\bar{k}+1} f(\mathfrak{S}(\mathbf{x}_{\bar{k}+1})) - \mu_{\bar{k}+1} g(\mathfrak{S}(\mathbf{x}_{\bar{k}+1})) \\ &\geq f(\mathfrak{S}(\mathbf{x}_{\bar{k}+1})) - \mu_{\bar{k}+1} L_f^2 - g(\mathfrak{S}(\mathbf{x}_{\bar{k}+1})) \\ &= F(\mathbf{x}_{\bar{k}+1}) - \mu_{\bar{k}+1} L_f^2, \end{aligned}$$

by a similar discussion in (34). Then, we have

$$\begin{aligned} \sum_{k=\underline{k}}^{\bar{k}} c\gamma_k \|\nabla F_k(\mathbf{x}_k)\|^2 &\leq F_1(\mathbf{x}_1) - F(\mathbf{x}_{\bar{k}+1}) + \mu_1 L_f^2 \\ &\leq F_1(\mathbf{x}_1) - \inf_{\mathbf{x} \in \mathbb{R}^d} F(\mathbf{x}) + \mu_1 L_f^2 < \infty, \end{aligned} \quad (37)$$

where $\inf_{\mathbf{x} \in \mathbb{R}^d} F(\mathbf{x}) > -\infty$ holds by Definition I.2 (c). Now, from Definition III.5 and Definition III.4 (a), we can bound γ_k from below, i.e., we have

$$\gamma_k \geq \bar{\delta} \kappa_{\mu_k}^{-1} := \min\{\delta, 2(1-c)\rho\} \kappa_{\mu_k}^{-1}.$$

From this inequality, we obtain

$$\begin{aligned} \sum_{k=\underline{k}}^{\bar{k}} c\gamma_k \|\nabla F_k(\mathbf{x}_k)\|^2 &\geq c\bar{\delta} \sum_{k=\underline{k}}^{\bar{k}} \kappa_{\mu_k}^{-1} \|\nabla F_k(\mathbf{x}_k)\|^2 \\ &= c\bar{\delta} \sum_{k=\underline{k}}^{\bar{k}} \frac{\mu_k}{\varpi_1 \mu_k + \varpi_2} \|\nabla F_k(\mathbf{x}_k)\|^2 \\ &\geq c\bar{\delta} \sum_{k=\underline{k}}^{\bar{k}} \frac{\mu_k}{\varpi_1 (2\eta)^{-1} + \varpi_2} \|\nabla F_k(\mathbf{x}_k)\|^2 \\ &\geq \frac{2\eta c\bar{\delta}}{\varpi_1 + 2\eta\varpi_2} \min_{\underline{k} \leq k \leq \bar{k}} \|\nabla F_k(\mathbf{x}_k)\|^2 \sum_{k=\underline{k}}^{\bar{k}} \mu_k, \end{aligned} \quad (38)$$

where we employed $\mu_k \leq (2\eta)^{-1}$ in the second inequality. Hence, (16) follows from (37) and (38) with

$$C := \frac{(\varpi_1 + 2\eta\varpi_2)(F_1(\mathbf{x}_1) - \inf_{\mathbf{x} \in \mathbb{R}^d} F(\mathbf{x}) + \mu_1 L_f^2)}{2\eta c\bar{\delta}} \in \mathbb{R}_{++}.$$

(b) From (13) (ii), we can get the following by taking the limit as $\bar{k} \rightarrow \infty$ on the both sides of (16):

$$\inf_{\underline{k} \leq k} \|\nabla F_k(\mathbf{x}_k)\| \leq 0.$$

Hence, (17) is obtained by taking the limit as $\underline{k} \rightarrow \infty$.

(c) From Definition III.7 (b), we can construct $(\mathbf{x}_{m(l)})_{l=1}^\infty$ such that $\lim_{l \rightarrow \infty} \|\nabla F_{m(l)}(\mathbf{x}_{m(l)})\| = 0$, e.g., in the same manner as in [45, footnote 11]. Definition III.1 (b) leads to $\text{dist}(\mathbf{0}, \partial_L(f \circ \mathfrak{S})(\bar{\mathbf{x}}) - \partial_L(g \circ \mathfrak{S})(\bar{\mathbf{x}})) = 0$ for any cluster point $\bar{\mathbf{x}}$ of $(\mathbf{x}_{m(l)})_{l=1}^\infty$, which completes the proof. $\square$

Supplementary Material for “A DC Composite Optimization via Variable Smoothing for Robust Phase Retrieval with Nonconvex Loss Functions”

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As described in Section IV, we conducted the experiments by using not only Cauchy outlier but also uniformly distributed outliers. In this material, we present the results for the uniformly distributed outliers for completeness.

Fig. S1 to S3 and Table S1 below correspond to Fig. 1 to 3 and Table II in Section IV, respectively. Despite the change in the distribution of ξ_i , each figure exhibits similar trends to those in the corresponding figure. Specifically, (i) Fig. S1 and S2 show the robustness of the proposed method with the capped ℓ_1 and the trimmed ℓ_1 , (ii) Fig. S3 illustrates that the appropriate values of β and K depend on the scale and number of outliers, respectively, and (iii) Table S1 demonstrates that the proposed method achieves lower computational time than the existing method.

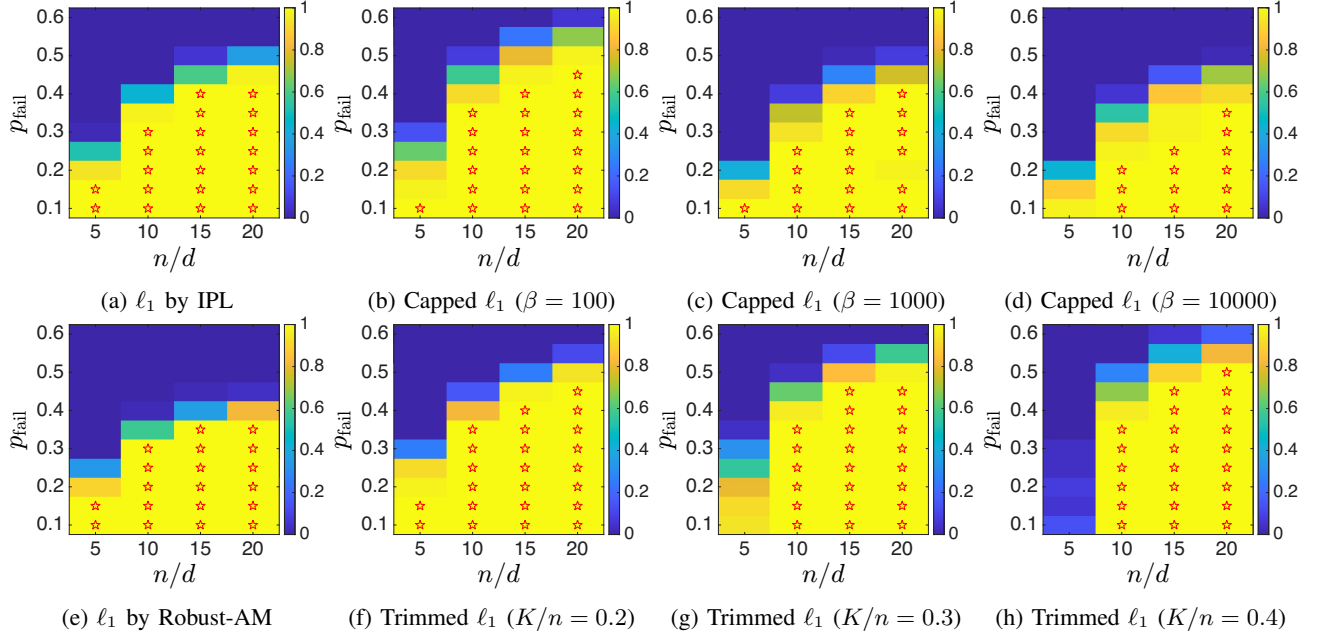


Fig. S1: Success rate of each method. We used uniformly distributed ξ_i and the same parameters $(d, s) = (100, 1)$ as in Fig. 1.

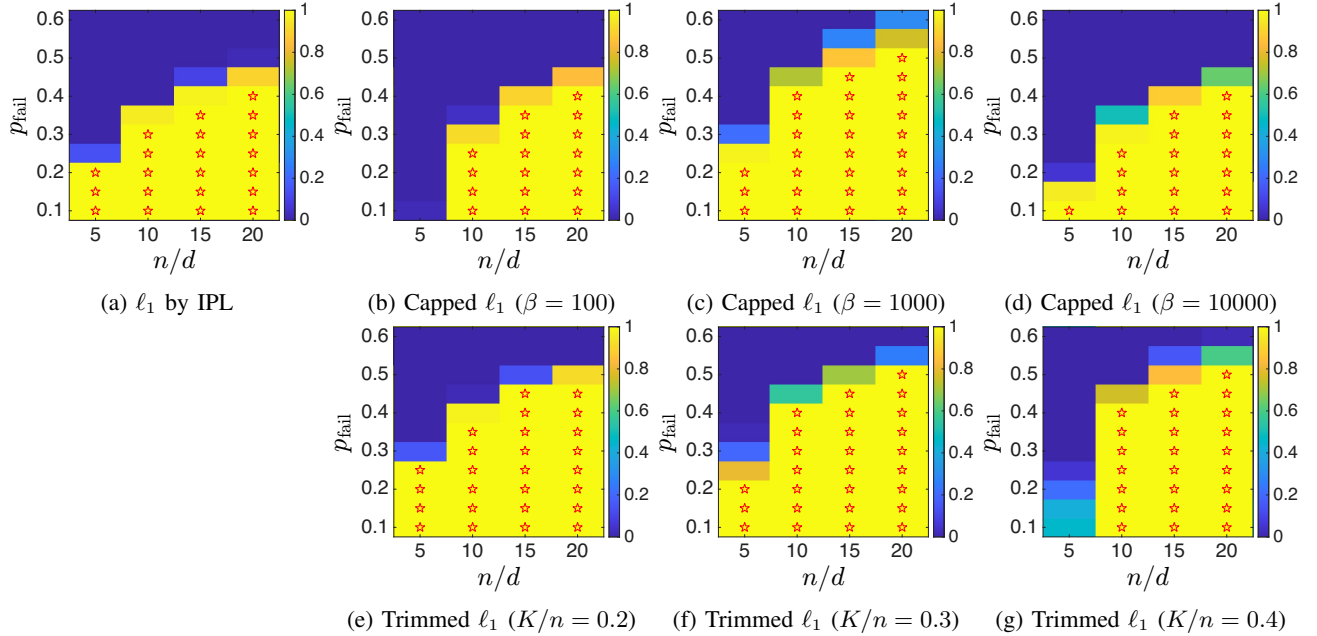


Fig. S2: Same as Fig. S1 except that $d = 500$. We omitted the result of Robust-AM because the subproblem solver (ADMM-LAD) did not reach its stopping criterion within 30 seconds even when $p_{\text{fail}} = 0.1$.

TABLE S1: The running CPU time in seconds (outside the parentheses) and the the number of iterations (inside the parentheses) for the case where ξ_i is uniformly distributed and $(d, p_{\text{fail}}, s) = (500, 0.35, 1)$.

n/d	5	10	15	20
ℓ_1 by IPL	30.00 (251.66)	7.14 (24.74)	7.95 (7.44)	10.13 (6.00)
Capped ℓ_1 ($\beta = 100$)	0.62 (258.26)	1.39 (313.78)	1.47 (242.10)	1.60 (213.88)
Capped ℓ_1 ($\beta = 1000$)	0.37 (150.86)	0.56 (118.94)	0.55 (82.28)	0.53 (63.14)
Capped ℓ_1 ($\beta = 10000$)	0.39 (158.62)	0.71 (154.98)	0.42 (59.80)	0.45 (51.16)
Trimmed ℓ_1 ($K/n = 0.2$)	1.44 (464.40)	0.82 (129.86)	0.93 (102.00)	1.03 (89.76)
Trimmed ℓ_1 ($K/n = 0.3$)	2.15 (698.48)	5.38 (969.08)	6.27 (781.86)	5.89 (595.94)
Trimmed ℓ_1 ($K/n = 0.4$)	1.87 (601.60)	1.86 (53.18)	1.82 (44.98)	4.20 (44.98)

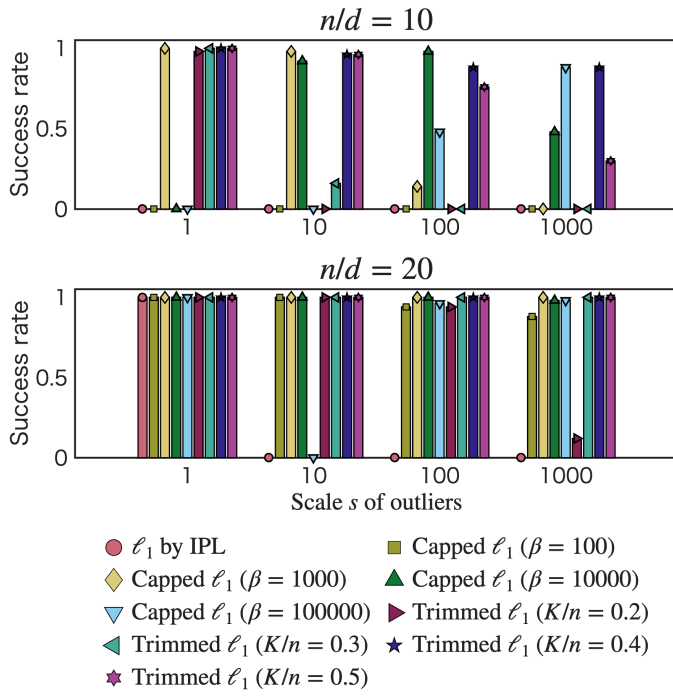


Fig. S3: Success rate for different values of s . We employed uniformly distributed ξ_i and the same parameters $(d, p_{\text{fail}}) = (500, 0.4)$ as in Fig. 3